

ENGLISH TEST

45 Minutes—75 Questions

DIRECTIONS: In the five passages that follow, certain words and phrases are underlined and numbered. In the right-hand column, you will find alternatives for each underlined part. You are to choose the one that best expresses the idea, makes the statement appropriate for standard written English, or is worded most consistently with the style and tone of the passage as a whole. If you think the original version is best, choose "NO CHANGE."

You will also find questions about a section of the passage, or about the passage as a whole. These questions

do not refer to an underlined portion of the passage, but rather are identified by a number or numbers in a box.

For each question, choose the alternative you consider best and fill in the corresponding oval on your answer document. Read each passage through once before you begin to answer the questions that accompany it. You cannot determine most answers without reading several sentences beyond the question. Be sure that you have read far enough ahead each time you choose an alternative.

PASSAGE I

The First Heart Surgeon

Daniel Hale Williams, an African-American physician, was the first person to perform successful heart surgery. His courage, as much as his skill, in meeting an emergency, enabled him. To save a life and¹ enter medical history.

Williams graduated from Chicago Medical College in 1883, being a revolutionary time for medicine. Because of² Louis Pasteur's discovery of the link between germs and disease, as a result of which Joseph Lister of England was³

advocating antiseptic use should there be during⁴ surgery. Before that time, many patients died from post-surgical infections. Williams adopted Lister's principals: he scrubbed the operating room with⁵ soap and water, sprayed it with germ killer, and sterilized his hands, his instruments, and his surgical gown. As a result, the mortality rate of Williams's surgical patients dropped at an alarming rate.⁶

1. A. NO CHANGE
B. meeting an emergency enabled him to
C. meeting an emergency. Enabled him to,
D. meeting, an emergency, enabled him to
2. F. NO CHANGE
G. 1883,
H. 1883, it was
J. 1883. Which was
3. A. NO CHANGE
B. disease,
C. disease;
D. disease, the result was that
4. F. NO CHANGE
G. antiseptics should be used,
H. the use of antiseptics
J. that the use of antiseptics
5. A. NO CHANGE
B. Listers principles:
C. Lister's principals:
D. Lister's principals in that
6. F. NO CHANGE
G. in its own terms.
H. alarmingly.
J. dramatically.

Despite such advances, most doctors were still afraid to perform chest surgery in spite of the⁷ risk of infection. But on July 10, 1893, in the emergency ward of Chicago's Provident Hospital, which he had founded in 1891, Williams faced a stark choice. Should he let a man die; or attempt an operation⁸ on his heart. The patient was James Cornish, a young man with a knife wound in his chest. Williams decided to operate. When he opened the chest cavity, he finds⁹ a cut in the pericardium (the sac surrounding the heart). Williams stitched up the cut and then began¹⁰ the vigil to see if his patient would survive.

Cornish did not develop an infection, and he was released from Provident in 1893 after fifty-one days,¹¹ completely recovered. Headlines worldwide acclaimed

Williams for his achievement. Given to the¹² limitations of medicine in 1893, his success

seemed—and still seem¹³—amazing. Without X rays, blood transfusions, or antibiotics, he

repaired a beating heart and led surgery into a new era.¹⁴

Cornish, the first heart-surgery patient, was

there in the operating room too, so it is ironic that his¹⁵ name is less well known than his doctor's.

7. A. NO CHANGE
B. due so greatly to
C. because of
D. because

8. F. NO CHANGE
G. choice: either let a man die, or
H. choice, either let a man die? Or
J. choice either let a man die or

9. A. NO CHANGE
B. is able to find
C. discovers
D. found

10. F. NO CHANGE
G. begun
H. had begun
J. had begun

11. A. NO CHANGE
B. after fifty-one days in 1893
C. after fifty-one days, in 1893,
D. in fifty-one days,

12. F. NO CHANGE
G. Given,
H. Given
J. Giving

13. A. NO CHANGE
B. seems
C. seemingly
D. do seem

14. F. NO CHANGE
G. led a new era into surgery and repaired a beating heart.
H. led surgery into a beating heart and repaired a new era.
J. repaired a beating heart surgery and led into a new era.

15. Which of the choices (all of them are accurate) should the writer use here to provide a fact that confirms Dr. Williams's success in treating James Cornish?
A. NO CHANGE
B. so he deserved to witness this historic medical development.
C. at the outset, and many later patients could only imagine what it must have been like for him.
D. to witness later progress: he is known to have lived at least twenty more years.

PASSAGE II

A Valuable Legacy

In the fourth millennium B.C., the

Sumerians, whose origins are unknown but whose legacy is otherwise, settled in the fertile plain of the

16

Tigris and Euphrates rivers. 17 They were the

first, to divide the day into twenty-four hours, the hour

18

into sixty minutes, and the circle into 360-degrees. They were also the first to leave a written record. Sumerian, 5,000 years old, is the earliest written language.

By 3200 B.C., Sumerian society had

become so increasing in size and complexity that a means

19

of keeping track of holdings and taxes became necessary.

Using clay tablets and sharpened reeds, the Sumerians made simplified drawings of what they inventoried. A drawing of an ox followed by two dots meant two oxen.

Scribes soon realized they could also stand for sounds.

20

Tl is the Sumerian word for arrow; the sign for an arrow

21

could represent both the object and the sound "*Tl*." This

realization made it possible to store even the most

22

complex information.

By 2500 B.C., scribes could write anything that could

16. The writer wants to emphasize that the Sumerians left a valuable legacy. Which choice does that best?

F. NO CHANGE
G. rich,
H. available,
J. accountable,

17. Which of the following sentences, if inserted here, would best illustrate the accomplishments of the Sumerians?

A. Early tribes often settled in fertile plains near rivers so they would have water for crops and transportation.
B. The Sumerians were a very accomplished group.
C. There, they devised an irrigation network, metal tools, and the wheel.
D. During their time, the Sumerians accomplished many very important things.

18. F. NO CHANGE

G. first to divide the day,
H. first to divide, the day
J. first to divide the day

19. A. NO CHANGE

B. so increased in size and complexity
C. become so excessively monolithic and circumfused by societal intricacies whose implications dictated
D. became so large and real hard to figure out

20. F. NO CHANGE

G. that they
H. these stylized pictures
J. that using them they

21. A. NO CHANGE

B. arrow! The
C. arrow! And the
D. arrow: and the,

22. F. NO CHANGE

G. evenly to
H. up evenly to
J. even for

be said. Using combinations of wedge-shaped marks that

came to be called cuneiform writing, they recorded

23

a lot of important details about their life—the entire spec-

24

trum of a culture. Among the thousands of tablets the

Sumerians left are the world's oldest medical prescription,

the oldest recipe (how to make beer), and even the oldest

school lesson. A clay tablet with rudimentary wedge forms

25

neatly imprinted on top by the teacher and awkwardly

copied by the student (students in the United States have

26

tablets to practice their writing on, too).

26

Cuneiform spread rapidly throughout the

Near East. It was adopted by the Akkadians, Hittites,

Persians, and others, and used for over 3,000 years.

27

Sumerian's scribes; created and regularly used at least 800

28

different cuneiform signs. Because it was cumbersome,

cuneiform was eventually replaced by alphabet-based

29

writing systems. 30

23. A. NO CHANGE

B. had came

C. have came

D. would have come

24. Given that all of them are true, which choice supports the paragraph by giving the most specific details?

F. NO CHANGE

G. not only business transactions, but fables, myths, poetry, proverbs—

H. in cuneiform a great many specific details about their lives—

J. many details about their businesses and culture—

25. A. NO CHANGE

B. lesson? A

C. lesson! A

D. lesson—a

26. F. NO CHANGE

G. student; American students practice writing, too.

H. student—all students need practice.

J. student.

27. A. NO CHANGE

B. using it

C. useful

D. that was used

28. F. NO CHANGE

G. Sumerians scribe's

H. Sumerian scribes

J. Sumerian scribe's

29. A. NO CHANGE

B. had been

C. has been

D. being

30. The writer wants the final sentence of the last paragraph to reflect the main idea of the whole essay. Given that all of the following sentences are true, which one, if inserted here, would do that best?

F. Even so, the ability to store and retrieve information without relying solely on human memory, from the first grader's laborious scrawl to sophisticated computers, is indebted to cuneiform.

G. Historians identify four main periods of Sumerian: Archaic Sumerian, Old or Classical Sumerian, New Sumerian, and Post-Sumerian.

H. Although Sumerian was replaced about 2000 B.C. as a spoken language, it continued in written use for many, many years.

J. Much of Sumerian literature remained in the oral tradition until the Old Babylonian period, when the literature was first put into writing.

Roommates

[1]

I don't know how the college dorm staff matched Portia and I as roommates.

31

They are supposed to used the freshman

32

dean's survey. It asked if you

33

smoke, you play music after midnight; and if you are very tidy, a little tidy, or an out-and-out slob. Both Portia and I considered ourselves

"a little tidy," and neither of us smoked. 35

[2]

I would liked playing music after midnight. I checked "no" on the survey only because I didn't have a stereo. Portia did *not* like to listen to music after midnight.

37 However, she did something much worse: she practiced the French horn.

[3]

I had a Southern accent; but to Portia's distress, little interest in the Civil War. Portia was a history buff. For

38

31. A. NO CHANGE

B. me as for

C. me as

D. I were

32. F. NO CHANGE

G. I suppose, they

H. They're supposedly

J. I suppose they

33. A. NO CHANGE

B. One was

C. It was

D. We were

34. F. NO CHANGE

G. smoke, then

H. smoke; if

J. smoke, do

35. Which of the following sentences, if added here, would most effectively signal the essay's shift in focus occurring at this point?

A. I was glad we had these things in common.

B. There the resemblance ended.

C. But that was just the beginning.

D. I knew it was going to be a great school year!

36. F. NO CHANGE

G. have like playing

H. like to of played

J. have liked to play

37. The writer had considered deleting the italics on the word *not* in the preceding sentence and revising "did not" to read as the contraction "didn't." If the writer had done this, the sentence would have lost its:

I. emphasis on the difference between the narrator's and Portia's preferences.

II. implication that Portia might be persuaded to change her mind on this matter.

III. suggestion that Portia has strong feelings about this matter.

A. I only

B. III only

C. I and II only

D. I and III only

38. F. NO CHANGE

G. accent but, to Portia's distress,

H. accent but, to Portia's distress

J. accent, but to Portias' distress

fun, she and her high school friends toured Civil War battle sites and pretended they were soldiers in the Union army.

[4]

I loved to talk about my family—about my dad's homemade peach cobbler or my brother's fascination with Welsh corgies. Portia acknowledged only that she had parents. She was intriguingly closemouthed on the topic of siblings. I learned, quite by happenstance, that Portia had at least one sibling: a fraternal twin, *who was in our freshman class!* When I confronted Portia, she told me that we were both sick and tired of being compared with ³⁹ her twin, who was a genius, and that I should just forget I ever knew this information.

[5]

[1] Like an unhappily married couple,

Portia and I carve out separate spheres in our twelve-by-⁴⁰

sixteen-foot space. [2] Her side, with pairs of shoes lined ⁴¹ up neatly on the windowsill, looked like my

grandmother's apartment. [3] My side, with a nursery of ⁴² ducks, looked like a mobile wearing ski caps. [4] I'd like to say that we became friends. [5] We didn't. [6] We didn't even consider rooming together sophomore year.

[7] We did see each other occasionally though, over the

39. A. NO CHANGE
B. she was not only
C. they both were
D. she was

40. F. NO CHANGE
G. are carving up
H. were carved out of
J. carved out

41. A. NO CHANGE
B. side with pairs of shoes,
C. side, with pairs of shoes,
D. side with pairs of shoes

42. F. NO CHANGE
G. with a mobile of ducks wearing ski caps, looked like a nursery.
H. looked like a nursery of ski cap-wearing ducks with a mobile.
J. with a nursery, looked like a mobile of ducks wearing ski caps.

summer: our chance meetings were always moderated by
an overwhelming sense of wariness. 44

43

43. The writer wishes to add a light note at this point, supporting the essay's sense of completion by tying the ending back to the essay's beginning. Given that all are true, which choice would best accomplish this?
- A. NO CHANGE
 - B. Portia had decided to spend the summer with her parents and twin sister.
 - C. we had both volunteered to help the freshman dean revise her survey.
 - D. it takes a great effort for roommates to become friends.
44. The writer is considering the division of Paragraph 5 into two separate paragraphs. In terms of the logic and coherence of the essay, the best course of action to take would be to:
- F. leave the paragraph as it is (NO CHANGE).
 - G. begin a new paragraph with Sentence 4.
 - H. begin a new paragraph with Sentence 5.
 - J. begin a new paragraph with Sentence 7.

PASSAGE IV

A Day Hike into the Jungle

[1]

We were hiking in the jungle on the island of Java, www.crackab.com
 which is part of the country of Indonesia. Surrounded
 by a dense growth of trees with long, dangling vines
and trunks, that rose a hundred feet into the air.

46

A flock of brightly colored parrots screeched at us, it
being that they were in the lower branches of a nearby
tree.

[2]

I was with an Indonesian friend, and she had offered
 to take me on a day hike into the jungle and show me a
 rare and unusual sight. She had refused, for example, to
 tell me what it was. My friend was a practical joker and
 liked to give me a shock now and again. Therefore, I was
 a little nervous and prepared for almost anything.

45. A. NO CHANGE
 B. Having been surrounded
 C. My companion and I was surrounded
 D. We were surrounded
46. F. NO CHANGE
 G. trunks that rose a hundred feet,
 H. trunks that rose a hundred feet
 J. trunks that rose, a hundred feet
47. A. NO CHANGE
 B. Being near the lower branches, a flock of brightly colored parrots screeched at us from the trees.
 C. Being that we were near the lower branches of a tree, a flock of brightly colored parrots screeched at us.
 D. A flock of brightly colored parrots screeched at us from the lower branches of a nearby tree.
48. F. NO CHANGE
 G. friend and
 H. friend,
 J. friend
49. A. NO CHANGE
 B. in other words,
 C. however,
 D. OMIT the underlined portion.
50. F. NO CHANGE
 G. Regardless,
 H. However,
 J. Then,

Following her, my friend chuckled at my apprehension.

51

[3]

We left the trail and crept slowly into the dense undergrowth. Then, I noticed the distinctive smell of rotting meat. My friend stopped. She carefully parted some bushes. She pointed at the ground. I was amazed. There in front of me was the largest flower I had ever seen. Its big, thick petals were bright scarlet, and they had spots on them that were white.

53

54 Altogether, it was nearly three feet wide.

[4]

It was definitely the largest and possibly, the prettiest but it was also the worst-smelling flower I'd ever encountered. The putrid stench of rotting flesh emanated from its oversized petals. We held our noses. "The flower's name is Rafflesia and its smell attracts insects," my friend explained. "Its pollen sticks to them and then they fly off to other plants. That's how the species is pollinated."

55

51. A. NO CHANGE
 B. Motioning for me, I followed my friend, who chuckled at my apprehension.
 C. Chuckling at my apprehension, a motion was made for me to follow her.
 D. Chuckling at my apprehension, my friend motioned for me to follow her.

52. F. NO CHANGE
 G. (Do NOT begin new paragraph) Pointing
 H. (Do NOT begin new paragraph) She pointed,
 J. (Begin new paragraph) She pointed

53. A. NO CHANGE
 B. spots of
 C. dotted with spots of
 D. OMIT the underlined portion.*

54. At this point, the writer would like to help readers visualize the flower by comparing it to things most readers are familiar with. Which of the following sentences, if added here, would most effectively accomplish this?
 F. It was too large to be supported by a stem; it was simply attached right to the ground.
 G. It looked like a cross between a monstrous rose and an upside-down church bell.
 H. It reminded me of a flower I had seen once in a children's book.
 J. It was almost a shame that something so beautiful was hidden so deep in a faraway jungle.

55. A. NO CHANGE
 B. largest and possibly, the prettiest,
 C. largest, and possibly, the prettiest
 D. largest, and possibly the prettiest,

[5]

I half believed this magnificent thing to jump up
 from its roots and run at me. But my friend only laughed.
 "Don't worry," she said. "It only smells like meat—it
 doesn't eat it." 57

56. F. NO CHANGE
 G. thought about expecting
 H. nearly almost expected
 J. think I half expected

57. The writer would like to add an introductory sentence to Paragraph 5 that shows why the writer's friend began to laugh. Which of the choices does that best?
- A. I wondered how she knew so much.
 B. I had never smelled anything so foul.
 C. Instinctively I backed away.
 D. My friend sure had surprised me.

Questions 58 and 59 ask about the essay as a whole.

58. The writer is considering adding the following sentence to the essay, in order to emphasize that she felt reluctant to enter the jungle:

Since I wasn't at all familiar with this part of the jungle, I had no choice but to stick close.

If added, this new sentence would best support and be placed after the last sentence in Paragraph:

- F. 2.
 G. 3.
 H. 4.
 J. 5.

59. Suppose the writer had been assigned to write an article for a travel magazine that described the various plants found in the jungles of Indonesia. Would this essay successfully fulfill the assignment?

- A. No, because the essay describes the writer's experience seeing an unusual species of flower in Indonesia.
 B. No, because the essay focuses on the jokes the writer plays on her friend and does not describe species of plants in detail.
 C. Yes, because the essay focuses on how plants pollinate their species in Indonesia.
 D. Yes, because the essay states that the jungle contained dense foliage.

PASSAGE V

My Favorite Place in the Park

Though thousands of tourists stream through
 Yellowstone National Park every summer, few ever visit
 my favorite place in the park. It is not spectacular like Old
 Faithful or Yellowstone Falls, where can you fly-fish there
 as one will on the Madison or Gallatin rivers. It is just a
 dirt field located in the woods behind the gift shops and
 cabins at Fishing Bridge, on the north shore of Lake
 Yellowstone, yet it brings around two of my great

60. Which choice provides the most appropriate image?

- F. NO CHANGE
 G. trickle
 H. ebb
 J. overflow

61. A. NO CHANGE

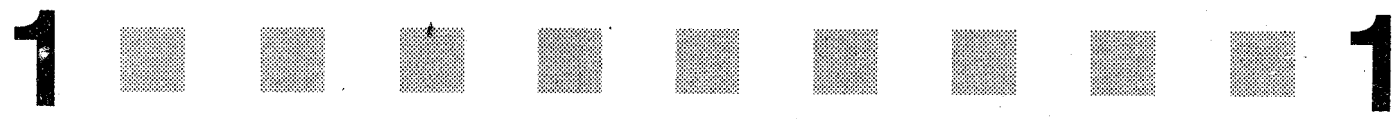
- B. and
 C. nor
 D. but

62. F. NO CHANGE

- G. they could
 H. you can
 J. tourists can

63. A. NO CHANGE

- B. they bring down
 C. it brings about
 D. it brings together



loves; the forest and baseball.

64

For two summers I worked as a breakfast-
and-lunch waiter in the restaurant at the Lake

Hotel, where it costs fifty dollars a night for a room,

65

and almost every day after work I would ride my bike to
the field to play ball. The field had been hewn out of the
dark stands of spruce, fir, and pine that now define it's
boundaries. A rickety fence barely separates the outfield

66

from the wilderness, and in right field a stream cuts away
a corner of the field. A foul ball requires a search-and-
rescue mission.

67

I remember standing in the outfield on one
of those hot summer days, lost from my thoughts.

68

A mosquito was poking at my forearm, and the batter

69

seemed to blur into the woods behind home plate and we
weren't playing to win, only to play. Yet, when I saw the
ball jump off the bat late in the game, visions of making
an over-the-shoulder catch flashed across my mind, and I
ran.

70

[1] There is something exhilarating about watching a
white ball spinning in an arc through a blue sky, your legs
flying and your glove stretched out as far as you can
reach. [2] When the dust had settled, I was staring at an
upside-down forest. [3] I leapt and caught the ball,
toppling over the fence and landing on my head. [4] A
large brown bird sat high on (or hung down from) a
branch. [5] I wondered if this wasn't some great owl
watching us, some avian sportscaster flapping its feathers

71

64. F. NO CHANGE
G. loves—the forest,
H. loves: the forest
J. loves, the forest,

65. A. NO CHANGE
B. Hotel,
C. Hotel. They charge fifty dollars for a room,
D. Hotel, where the rooms cost fifty dollars a night,

66. F. NO CHANGE
G. its
H. their
J. its'

67. A. NO CHANGE
B. wilderness, then
C. wilderness, which
D. wilderness

68. F. NO CHANGE
G. loss to
H. lost in
J. lost of

69. A. NO CHANGE
B. My forearm, being poked at by a mosquito, and
C. When a mosquito was poking at my forearm,
D. A mosquito poking at my forearm and

70. F. NO CHANGE
G. plate. We
H. plate so we
J. plate because we

71. A. NO CHANGE
B. large, brown bird,
C. large brown bird,
D. large, brown, bird

over our ball game, at least 500 miles from the nearest
big-league ballpark. 73

72

72

72. The writer wishes to conclude this sentence with a phrase that would continue its analogy and be consistent with the whimsical tone of the reminiscence. Which choice would best accomplish this?
- F. NO CHANGE
 - G. announcing who was at bat to the surrounding forest.
 - H. like it had never seen such a spectacle before.
 - J. the only one being played in Yellowstone Park.
73. For the sake of unity and coherence, Sentence 2 of the last paragraph should be placed:
- A. where it is now.
 - B. immediately before Sentence 1.
 - C. immediately before Sentence 4.
 - D. immediately before Sentence 5.

Questions 74 and 75 ask about the essay as a whole.

74. The writer is considering the addition of the following sentence to the essay:

Established in 1872 as the first national park in the United States, Yellowstone boasts a native population of buffalo, elk, deer, moose, bears, and coyotes.

Given that this statement is true, should it be added to the essay, and if so, where?

- F. Yes, at the beginning of the first paragraph, because the sentence effectively introduces the subject of this essay.
- G. Yes, at the beginning of the second paragraph, because the sentence refers to the animals that live in the forest.
- H. No, because such general information about Yellowstone would be inappropriate anywhere in this personal essay.
- J. No, because the account being related by the essay's narrator occurred much more recently than 1872.

75. Suppose that the editor of a magazine had assigned the writer to depict a firsthand adventure experienced while backpacking in the wilderness of Yellowstone National Park. Does the essay successfully fulfill this assignment?

- A. Yes, because the essay describes a secluded location in the park far from tourists.
- B. Yes, because the essay describes an outdoor sport being played in the wilds of Yellowstone.
- C. No, because the essay describes playing a game in Yellowstone rather than backpacking in it.
- D. No, because the essay's tone is too formal and too personal for such an assignment.

END OF TEST 1

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.



MATHEMATICS TEST

60 Minutes—60 Questions

DIRECTIONS: Solve each problem, choose the correct answer, and then fill in the corresponding oval on your answer document.

Do not linger over problems that take too much time. Solve as many as you can; then return to the others in the time you have left for this test.

You are permitted to use a calculator on this test. You may use your calculator for any problems you choose,

but some of the problems may best be done without using a calculator.

Note: Unless otherwise stated, all of the following should be assumed.

1. Illustrative figures are NOT necessarily drawn to scale.
2. Geometric figures lie in a plane.
3. The word *line* indicates a straight line.
4. The word *average* indicates arithmetic mean.

1. If $2x - 3 = 12$, then $x = ?$

A. 2.0
B. 4.5
C. 7.5
D. 9.0
E. 15.0

DO YOUR FIGURING HERE.

2. Consider the following two logical statements.

If the length of $\overline{AB}$ is 2, then the length of $\overline{BC}$ is 5.
The length of $\overline{BC}$ is NOT 5.

If these statements are both true, then it follows that the length of:

F. $\overline{AB}$ is NOT 2.
G. $\overline{AB}$ is 2.
H. $\overline{AB}$ is 5.
J. $\overline{BC}$ is 2.
K. $\overline{BC}$ is NOT 2.

3. If the probability that it will rain tomorrow is 0.7, what is the probability that it will NOT rain tomorrow?

A. 0.0
B. 0.1
C. 0.3
D. 1.0
E. 1.7



4. Mario bought 1 compact disc for \$8.95 and 3 others for \$7.99 each. What was the average price per disc he paid for these 4 compact discs?

F. $\$8.95 + \frac{\$7.99}{3}$

G. $\frac{\$8.95}{4} + \frac{\$7.99}{3}$

H. $\frac{\$8.95 + \$7.99}{2}$

J. $\frac{\$8.95 + \$7.99}{4}$

K. $\frac{\$8.95 + 3(\$7.99)}{4}$

DO YOUR FIGURING HERE.

5. On Saturday Sid received his pay and spent $\frac{1}{2}$ of it.

On Sunday he spent $\frac{1}{2}$ of the remaining money, and

on Monday he spent $\frac{1}{2}$ of what remained from

Sunday. If \$4 then remained, how much pay did he

receive originally?

A. \$12

B. \$16

C. \$24

D. \$32

E. \$64

6. If $N = -3a$ and $M = 2b - a$, then what is the value of $N + M$?

F. $-4a + 2b$

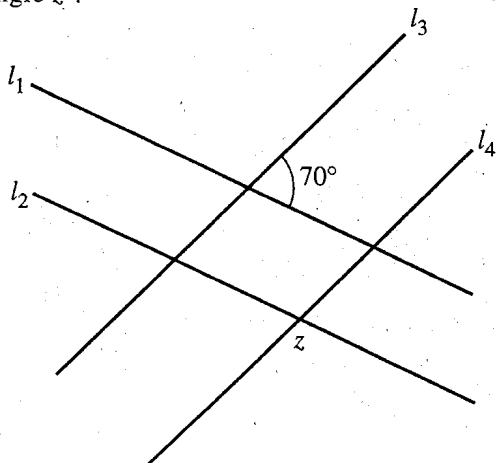
G. $-4a - 2b$

H. $-3a + 2b$

J. $-2a + 2b$

K. $-2a - 2b$

7. In the figure below, l_1 is parallel to l_2 , l_3 is parallel to l_4 , and the lines intersect as shown. What is the measure of angle z ?



A. 110°

B. 120°

C. 130°

D. 140°

E. Cannot be determined from the given information



8. If $x = -4$, then $-x^2 - 5x + 3 = ?$

- F. 31
- G. 15
- H. 7
- J. -25
- K. -33

DO YOUR FIGURING HERE.

9. The average of 6 numbers is 4.5. If each of the numbers is decreased by 4, what is the average of the 6 new numbers?

- A. 0.0
- B. 0.5
- C. 4.5
- D. 5.0
- E. 9.0

10. The expression $7a + 7b$ is equivalent to which of the following?

- F. $7(a + b)$
- G. $14(a + b)$
- H. $7a + b$
- J. $7ab$
- K. $14ab$

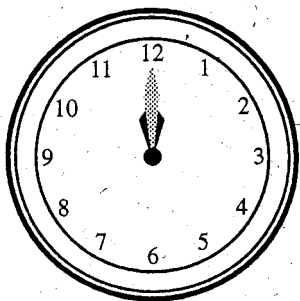
11. For each day on your newspaper route you receive \$10.00 plus a fixed amount for each newspaper you deliver. Currently you are earning \$18.00 per day for delivery of 80 newspapers. Today you are assigned to deliver 20 additional newspapers per day. What will be your new daily earnings?

- A. \$10.00
- B. \$18.20
- C. \$19.00
- D. \$20.00
- E. \$22.50

12. If $\frac{5}{x} \geq \frac{1}{3}$, what is the largest possible value for x ?

- F. $\frac{1}{2}$
- G. 2
- H. 3
- J. 8
- K. 15

13. What is the number of degrees the *hour* hand of a clock moves in 10 hours?



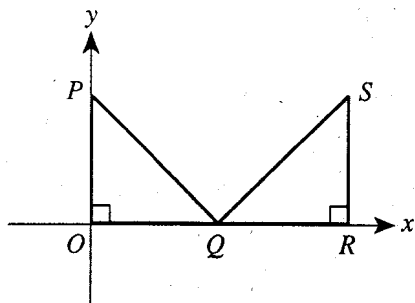
- A. 100°
- B. 120°
- C. 150°
- D. 220°
- E. 300°



14. In the standard (x,y) coordinate plane below, $\triangle POQ$ and $\triangle SRQ$ are isosceles right triangles with equal areas. Points O , P , Q , and R are located on the axes as shown. Which of the following could be the coordinates of point S ?

DO YOUR FIGURING HERE.

- F. $(0, 14)$
 G. $(7, 0)$
 H. $(7, 14)$
 J. $(14, 0)$
 K. $(14, 7)$



15. Which of the following equations has both $x = 2$ and $x = -3$ as solutions?

- A. $(x - 2)(x + 3) = 0$
 B. $(x + 2)(x + 3) = 0$
 C. $(x - 2)(-x + 3) = 0$
 D. $(x + 2)(x - 3) = 0$
 E. $x - 2 = x + 3$

16. A theater has m rows of seats with $(n + p)$ seats in each row. Which of the following is an expression for the number of seats in the entire theater?

- F. $(m \cdot n) + p$
 G. $(m \cdot n) + (m \cdot p)$
 H. $n + (m \cdot p)$
 J. $m \cdot n \cdot p$
 K. $m + n + p$

17. If 40% of x equals 80, then $x = ?$

- A. 2
 B. 32
 C. 200
 D. 2,000
 E. 3,200

18. The cost of electricity at the local utility company is 10 cents per kilowatt-hour for the first 100 kilowatt-hours used in a month and 7 cents per kilowatt-hour for each additional kilowatt-hour used. What is the cost for 500 kilowatt-hours used in 1 month?

- F. \$28.00
 G. \$35.00
 H. \$38.00
 J. \$50.00
 K. \$85.00



19. You are standing in line at the cash register to pay for a bracelet priced at \$15.99. A sales tax of 8% of the \$15.99 will be added (rounded to the nearest cent) to the price of the bracelet. You have 20 one-dollar bills, but how much will you need in coins if you want to have exact change ready?

DO YOUR FIGURING HERE.

- A. 7¢
 B. 8¢
 C. 19¢
 D. 27¢
 E. 63¢
20. For which nonnegative value of x is the expression

$$\frac{1}{9-x^2}$$

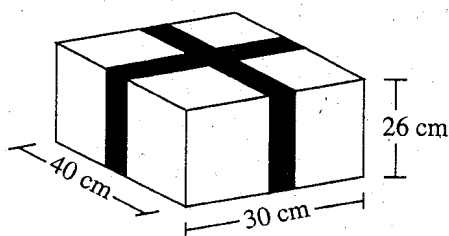
undefined?

- F. 81
 G. 18
 H. 9
 J. 3
 K. 0

21. What is the correct ordering of π , $\frac{7}{2}$, and 3 from least to greatest?

- A. $3 < \pi < \frac{7}{2}$
 B. $3 < \frac{7}{2} < \pi$
 C. $\pi < 3 < \frac{7}{2}$
 D. $\frac{7}{2} < 3 < \pi$
 E. $\frac{7}{2} < \pi < 3$

22. Two strips of tape are to be used to wrap a box, as shown below. Both strips must go completely around the box. What is the minimum length of tape, in centimeters (cm), required to wrap the box?



- F. 102
 G. 112
 H. 144
 J. 204
 K. 244

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23. You have been asked to make punch for a wedding reception. The punch recipe calls for 6 parts fruit juices to 2 parts soda. To make 24 quarts of this punch, how many quarts of soda should you use?

A. 3
B. 5
C. 6
D. 12
E. 20

DO YOUR FIGURING HERE.

24. Which of the following gives all the solutions of $x^2 - x = 12$?

F. -4 and 3
G. -3 and 4
H. -2 and 6
J. 12 and 13
K. 13 only

25. If $(m - n)^2 = 25$ and $mn = 36$, then $m^2 + n^2 =$?

A. -47
B. -11
C. 61
D. 86
E. 97

26. If, for all x , $(x^{2a-1})^3 = x^9$, then $a =$?

F. 1
G. $\frac{7}{5}$
H. $\frac{5}{3}$
J. 2
K. $\frac{12}{5}$

27. For the complex number i such that $i^2 = -1$, what is the value of $i^4 + 2i^2$?

A. -2
B. -1
C. 0
D. 1
E. 2

28. One endpoint of a line segment in the (x,y) coordinate plane has coordinates $(5,-8)$. The midpoint of the segment has coordinates $(-1,4)$. What are the coordinates of the other endpoint of the segment?

F. $(-7,16)$
G. $(-4,-32)$
H. $(2,-2)$
J. $(3,-6)$
K. $(4,12)$

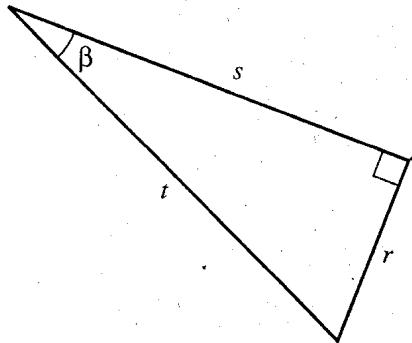


29. In the (x,y) coordinate plane, what is the radius of the circle $(x - 2)^2 + (y + 1)^2 = 6$?

DO YOUR FIGURING HERE.

- A. 1
B. $\sqrt{2}$
C. 2
D. $\sqrt{6}$
E. 6

30. In the right triangle pictured below, r , s , and t are the lengths of its sides. What is the value of $\cos \beta$?

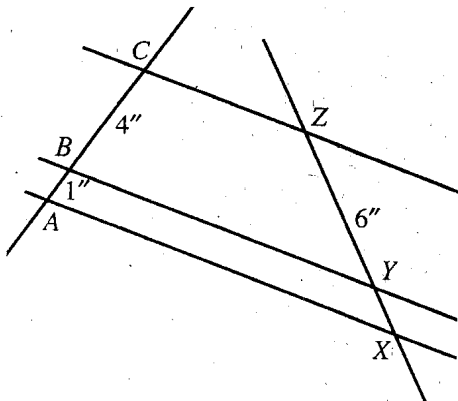


- F. $\frac{s}{r}$
G. $\frac{t}{r}$
H. $\frac{t}{s}$
J. $\frac{r}{t}$
K. $\frac{s}{t}$

31. For all nonzero a and b , $\frac{(10a^2b^2)(-9a^2b^3)}{6a^2b^4} = ?$

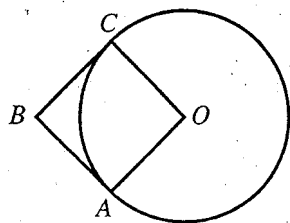
- A. $-15b$
B. $-15a^2b$
C. $-15a^2b^2$
D. $\frac{a^2b^2}{15}$
E. $\frac{12}{b}$

32. In the figure below, 3 parallel lines are crossed by 2 transversals as shown. The points of intersection and some distances, in inches, are labeled. What is the length, in inches, of $\overline{YX}$?



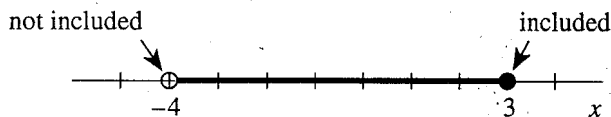
- F. 1
G. 2
H. 3
J. $\frac{3}{2}$
K. $\frac{5}{2}$

33. The figure below shows square $OABC$ and also shows the circle centered at O with radii $\overline{OC}$ and $\overline{OA}$. If the area of the square is 36 square units, what is the area of the circle, in square units?



- A. 12π
B. 18π
C. 36π
D. 72π
E. 81π

34. Which of the following logical statements identifies the same set as the graph shown below?



- F. $x \leq -4$ or $x > 3$
G. $x > -4$ or $x \leq 3$
H. $x \geq -4$ or $x < 3$
J. $x > -4$ and $x \leq 3$
K. $x \geq -4$ and $x < 3$

35. A forest begins 300 feet from the center of the base of a vertical television tower, across level ground. If the tower is 400 feet tall, how many feet is it from the top of the tower to the nearest edge of the forest at ground level?

- A. $100\sqrt{7}$
B. $200\sqrt{3}$
C. 500
D. 700
E. 120,000

36. If x and y are real and $\sqrt{2\left(\frac{x^2}{y}\right)} = 1$, then what *must* be true of the value of y ?

- F. y must be positive
G. y must be negative
H. y must equal $\frac{1}{2}$
J. y must equal 2
K. y may have any value

37. If c is a positive integer that divides both 75 and 90, but divides neither 18 nor 10, what should you get when you add the digits in c ?

- A. 1
B. 5
C. 6
D. 7
E. 8

DO YOUR FIGURING HERE.



38. What is the slope of any line parallel to the x -axis in the (x,y) coordinate plane?

DO YOUR FIGURING HERE.

- F. -1
 G. 0
 H. 1
 J. Undefined
 K. Cannot be determined from the given information

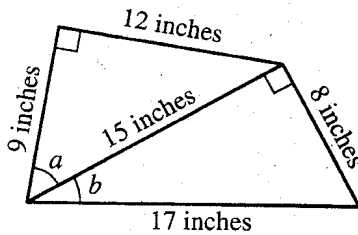
39. Which of the following lines has the largest slope?

- A. $y = 3x - 1$
 B. $y = 2x + 12$
 C. $y = 4x + \frac{1}{2}$
 D. $4y = 2x + 8$
 E. $5y = 10x - 2$

40. The 2 triangles in the figure below share a common side. What is $\sin(a + b)$?

(Note: $\sin(a + b) = \sin a \cos b + \sin b \cos a$ for all a and b .)

- F. $\frac{9}{17}$
 G. $\frac{20}{17}$
 H. $\frac{231}{255}$
 J. $\frac{252}{255}$
 K. $\frac{324}{255}$



41. Rita can walk 2 miles in m minutes. At that pace, how many minutes would it take her to walk 7 miles?

- A. $14m$
 B. $\frac{2}{7m}$
 C. $\frac{7}{2m}$
 D. $\frac{2m}{7}$
 E. $\frac{7m}{2}$

42. Which of the following calculations will yield an even integer for any integer n ?

- F. n^2
 G. $3n^2$
 H. $4n^2$
 J. $3n^2 + 1$
 K. $4n^2 + 1$

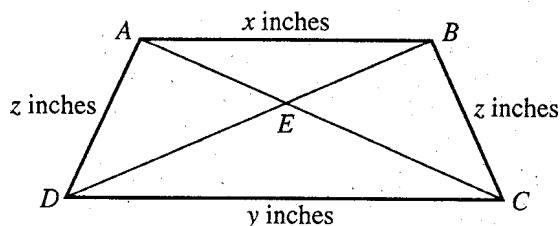


43. In $\triangle ABC$, the measure of $\angle A$ is 30° and the measure of $\angle B$ is 60° . If $\overline{AB}$ is 12 units long, what is the area, in square units, of $\triangle ABC$?

A. 12
 B. $18\sqrt{3}$
 C. 36
 D. 72
 E. $72\sqrt{3}$

DO YOUR FIGURING HERE.

44. Trapezoid $ABCD$ below is isosceles, with side lengths as marked. Its diagonals intersect at E . What is the ratio of the length of $\overline{AE}$ to the length of $\overline{DE}$?



F. $\frac{x}{y}$
 G. $\frac{x}{z}$
 H. 1
 J. $\frac{y}{z}$
 K. $\frac{z}{y}$

45. A certain rectangle is 3 times as long as it is wide. Suppose the length and width are both doubled. The area of the second rectangle is how many times as large as the area of the first?

A. 2
 B. 3
 C. 4
 D. 6
 E. 12

46. For what value of a would the following system of equations have an infinite number of solutions?

$$\begin{aligned} 2x - 3y &= 8 \\ 6x - 9y &= 4a \end{aligned}$$

F. 2
 G. 6
 H. 8
 J. 24
 K. 32

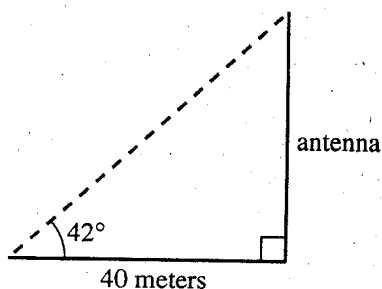


DO YOUR FIGURING HERE.

47. Two ships leave a port at the same time. One ship moves at 20 miles per hour and sails 2 hours north, and then 1 hour east. The other ship moves at 25 miles per hour and sails 2 hours east, and then 1 hour north. Which of the following is an expression for the number of miles apart the ships are 3 hours after they leave port?

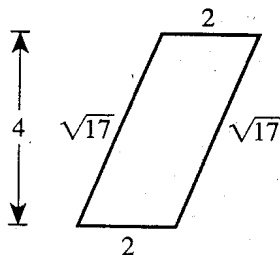
- A. $3(25 - 20)$
- B. $\sqrt{(50 - 20)^2 + (25 - 40)^2}$
- C. $\sqrt{(50 - 20)^2 + (25 + 40)^2}$
- D. $\sqrt{(50 + 20)^2 (25 - 40)^2}$
- E. $\sqrt{(3 \times 20)^2 + (3 \times 25)^2}$

48. When measured from a point on the ground that is 40 meters from the base of an antenna, the angle of elevation to the top of the antenna is 42° , as shown below. What is the height, in meters, of the antenna?



- F. $40 \cos 42^\circ$
- G. $40 \cot 42^\circ$
- H. $40 \sec 42^\circ$
- J. $40 \sin 42^\circ$
- K. $40 \tan 42^\circ$

49. In the parallelogram below, lengths are given in centimeters. What is the area of the parallelogram, in square centimeters?



- A. 4
- B. 8
- C. $\sqrt{17}$
- D. $\sqrt{34}$
- E. $2\sqrt{17}$

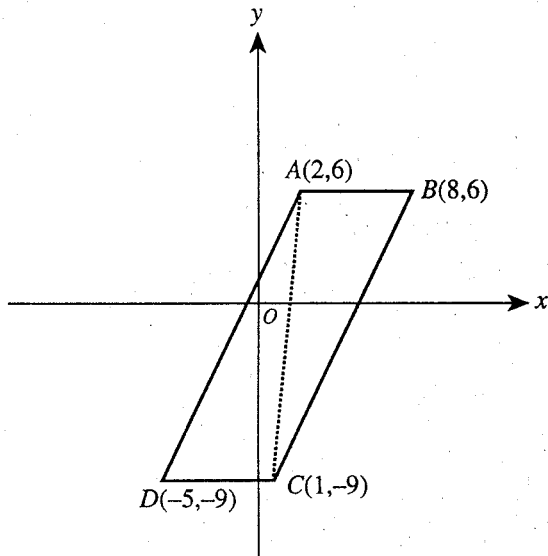


50. Points X , Y , and Z lie on the same line. If the length of $\overline{XY}$ is 5 meters and the length of $\overline{YZ}$ is 3 meters, then what are all the possible lengths, in meters, for $\overline{XZ}$?

F. 2 only
 G. 8 only
 H. 2 and 8 only
 J. Any number less than 2 or greater than 8
 K. Any number greater than 2 and less than 8

DO YOUR FIGURING HERE.

51. Given the vertices of parallelogram $ABCD$ in the standard (x,y) coordinate plane below,



what is the area of $\triangle ABC$, in square units?

A. 10.5
 B. 21.0
 C. 42.0
 D. 45.0
 E. 90.0

52. If $2a^2b^3 < 0$, then which of the following CANNOT be true?

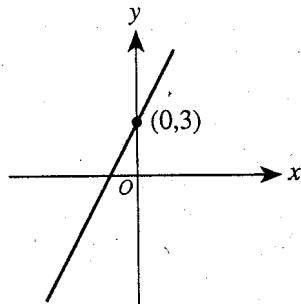
F. $a = b$
 G. $a < 0$
 H. $a > 0$
 J. $b < 0$
 K. $b > 0$

53. The first and second terms of a geometric sequence are n and an , in that order. What is the 1,000th term of the sequence?

A. $a^{999}n$
 B. $a^{1,000}n$
 C. $a^{1,001}n$
 D. $(an)^{999}$
 E. $(an)^{1,000}$



54. If a system of 2 linear equations in 2 variables has NO solution, and 1 of the equations is graphed in the (x,y) coordinate plane below, which of the following could be the equation of the other line?



- F. $y = -2x + 2$
 G. $y = 2x + 2$
 H. $y = -2x - 3$
 J. $y = 2x + 3$
 K. $y = 2$

55. If $0^\circ \leq x \leq 90^\circ$ and $\sin x = \frac{7}{25}$, then $\tan x = ?$

- A. $\frac{7}{24}$
 B. $\frac{24}{25}$
 C. $\frac{25}{24}$
 D. $\frac{24}{7}$
 E. $\frac{25}{7}$

56. For every cent decrease in price of a gallon of gasoline, a station sells 250 more gallons per day. The station normally sells 3,000 gallons of gasoline a day at \$1.09 per gallon. Which of the following expressions represents the number of gallons of gasoline sold per day if the cost is reduced by x cents per gallon?

- F. $3,000(1.09 - x)$
 G. $1.09(250x + 3,000)$
 H. $(1.09 - x)(250x + 3,000)$
 J. $1.09 - x$
 K. $3,000 + 250x$

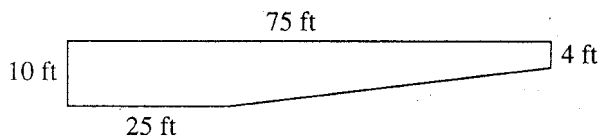
57. In a game, 100 marbles numbered 00 through 99 are placed in a box. A player draws 1 marble at random from the box. Without replacing the first marble, the player draws a second marble at random. If both marbles drawn have the same tens digit (that is, both are numbered between 00 and 09, or 10 and 19, or 20 and 29, etc.), the player is a winner. If the first marble Terry draws is numbered 87, what is the probability that Terry will be a winner on the next draw?

- A. $\frac{1}{8}$
 B. $\frac{1}{9}$
 C. $\frac{1}{10}$
 D. $\frac{1}{11}$
 E. $\frac{1}{99}$

DO YOUR FIGURING HERE.

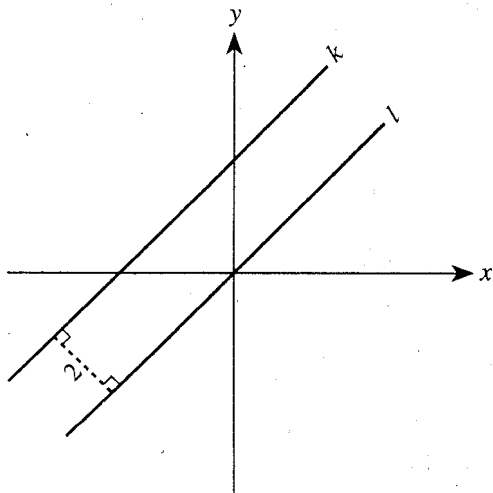


58. The figure below shows a cross section of a swimming pool that is 50 feet (ft) wide and 75 ft long. The bottom is horizontal for 25 ft, then slopes up, and the sides are vertical. What is the volume of water, in cubic ft, that is required to fill the pool to a depth, as shown, of 10 ft at the deep end and 4 ft at the shallow end?



- F. 600
G. 25,000
H. 26,250
J. 30,000
K. 47,500

59. In the figure below, line l has the equation $y = x$. Line k is above l , as shown, and k is parallel to l . Which of the following is an equation for k ?



- A. $y = x + 2$
B. $y = x + 2\sqrt{2}$
C. $y = x + \sqrt{3}$
D. $y = 2x$
E. $y = 2x + 2$
60. What is the smallest possible value for the product of 2 real numbers that differ by 6?

- F. -36
G. -9
H. -6
J. 0
K. 9

DO YOUR FIGURING HERE.

END OF TEST 2

READING TEST

35 Minutes—40 Questions

DIRECTIONS: There are four passages in this test. Each passage is followed by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

Passage I

PROSE FICTION: This passage is adapted from a short story by R. A. Sasaki in *The Loom and Other Stories* (©1991 by R. A. Sasaki).

She had grown up in San Francisco, wearing the two faces of a second-generation child born of immigrant parents. The two faces never met; there was no common thread running through both worlds. The duality was unplanned, untaught. Perhaps it had begun the first day of school when she couldn't understand the teacher and Eleanor Leland had called her a "Jap" and she cried. Before then there had never been a need to sort out her identity; she had met life headlong and with the confidence of a child.

Her world had been the old Victorian flat in which her mother took in boarders—the long, narrow corridor, the spiral stairway, the quilts covered with bright Japanese cloth, and the smell of fish cooking in the kitchen. She had accepted without question the people who padded in and out of her world on stockinged feet; they all seemed to be friends of the family. She never wondered why most of them were men, and it never occurred to her child's mind to ask why they didn't have their own families. The men often couldn't pay, but they were always grateful. They lounged in doorways and had teasing affectionate words for her and her sister. Then they would disappear, for a month, for six months, a year. Time, to a child, was boundless and unmeasurable. Later, crates of fruit would arrive and be stacked in the corridor. "From Sato-san," her mother would say, or "*Kudoh-san kara*."

The young men sometimes came back to visit with new hats set jauntily on their heads, if luck was good. But often luck was not good, and they came back to stay, again and again, each stay longer than the last; and each time they would tease less and drink more with her father in the back room. The slap of cards rose over the low mumble of their longing and despair. All this she accepted as her world.

The Victorian house which contained her world was on Pine Street, and so it was known as "Pine" to the young adventurers from her parents' native Wakayama prefecture in Japan who made their way from the docks of Osaka to the lettuce fields and fruit orchards of California. "Stay at Pine," the word passed along the grapevine, "Moriwaki-san will take care of you."

It was a short walk down the Buchanan Street hill from Pine to the flats where the Japanese community had taken root and was thriving like a tree whose seed had blown in from the Pacific and had held fast in this nook, this fold in the city's many gradations. When she was a little older her world expanded beyond the Victorian called Pine. It expanded toward the heart of this community, toward the little shops from which her mother returned each day, string bag bulging with newspaper-wrapped parcels. . . .

When it came time for her to go to school, she was not sent to the same school as the other Japanese-American children because Pine was on the edge of Japantown and in a different school district. She was the only Japanese in her class. . . .

She did her best to blend in. Though separated from the others by her features and her native tongue, she tried to be as inconspicuous as possible. If she didn't understand what the teacher said, she watched the other children and copied them. She listened carefully to the teacher and didn't do anything that might provoke criticism. If she couldn't be outstanding she at least wanted to be invisible.

She succeeded. She muted her colors and blended in. She was a quiet student and the other children got used to her; some were even nice to her. But she was still not really a part of their world because she was not really herself.

At the end of each school day she went home to the dark, narrow corridors of the old Victorian and the soothing, unconscious jumble of two tongues that was the two generations' compromise for the sake of communication. Theirs was a comfortable language, like a comfortable old sweater that had been well washed and rendered shapeless by wear. She would never wear it outside of the house. . . .

In the outside world—the *hakujin* world—there was a watchdog at work who rigorously edited out Japanese words and mannerisms when she spoke. Her words became formal, carefully chosen and somewhat artificial. She never thought they conveyed what she really felt, what she really was, because what she really was was unacceptable. In the realm of behavior, the watchdog was a tyrant. Respectability, as defined by

popular novels and Hollywood heroines, must be upheld at all costs. . . . She could admit to no weakness, no peculiarity. She would be irreproachable. She would be American.

1. One of the points suggested by lines 67–71 regarding the main character's learning to get along at school is that:
 - A. school enabled her to understand her inner self.
 - B. going to school made her feel like other kids.
 - C. doing so meant she could not be her whole true self.
 - D. blending in is the worst thing to do in a new setting.
2. As it is used in line 78, the word *rendered* most nearly means:
 - F. made.
 - G. seen as.
 - H. meant to be.
 - J. cleaned.
3. The young men who were boarders on Pine Street typically came back there to live because they:
 - A. had their ambitions frustrated by bad fortune.
 - B. deserved better than to harvest fruit and vegetables.
 - C. had grown tired of working on the docks in Osaka.
 - D. liked playing cards in the back room.
4. It can reasonably be inferred from lines 72–79 that the compromise made by the main character's parents was that they would:
 - F. stay in the Victorian house.
 - G. edit out their Japanese mannerisms.
 - H. be as inconspicuous as possible.
 - J. try to communicate in both languages.
5. Which of the following are included in the author's description of the old Victorian flat?
 - I. The smell of fish cooking
 - II. People coming and going
 - III. Unappreciative boarders
 - IV. Fruit stacked in the corridor
 - A. I and II only
 - B. II and IV only
 - C. I, II, and III only
 - D. I, II, and IV only
6. The passage claims that the "watchdog at work" (line 81) in the outside world caused the main character's speech to be:
 - I. conversational.
 - II. restricted.
 - III. unnatural.
 - F. II only
 - G. I and II only
 - H. II and III only
 - J. I, II, and III
7. The name of one of the main character's parents is:
 - A. Sato-san.
 - B. Moriwaki-san.
 - C. Kudoh-san kara.
 - D. not mentioned in the passage.
8. The main character's exhibiting respectability, as it is defined by the last paragraph, might mean all of the following EXCEPT:
 - F. being beyond reproach.
 - G. acting American.
 - H. imitating a Hollywood star.
 - J. speaking her mind.
9. The world of the old Victorian flat, as compared to the *hakujin* world, is judged in this passage to be a place:
 - A. that was more truly American.
 - B. where the boarders attempted to earn their fortunes.
 - C. made hospitable by the presence of many friends.
 - D. in which people's career ambitions were at risk.
10. The house in which the main character is brought up is located:
 - F. in the flats.
 - G. on a hill.
 - H. near the docks.
 - J. far from the *hakujin* world.

Passage II

SOCIAL SCIENCE: This passage is adapted from Reay Tannahill's *Food in History* (©1973 by Reay Tannahill).

The fields of wild grain which sprang up in the Near East twelve thousand years ago did more than help to feed humankind. They also attracted a number of those smaller animals which had begun to multiply in the open shade around the margins of the forests. In the early part of the growing season the raids made by wild goats and sheep on the new grain sprouting in the fields must have been a serious threat to the villagers' future food supply. They had three options—to exterminate the flocks, to defend the fields, or to bring the animals under control (which had the overriding advantage of ensuring the meat as well as the grain supply).

Domesticating the sheep and the goat may have been a comparatively easy task. Both were gregarious; where one led, the others would follow. In some villages it may have been commonplace to adopt orphan baby animals as pets, and a hand-reared ewe allowed to mate with a wild sire would, in the course of a few years, help to establish a village flock.

Whether the sheep or the goat was the first animal to be domesticated remains an open question, though the balance of probabilities favors the goat—an animal which has been greatly maligned through the ages, for its destructive browsing habits as well as its pungent smell. But during the early expansion of farming, in areas where scrub had to be cleared to make way for cultivation, its talent for killing plants by defoliation may have made the goat a useful agricultural laborer.

In the case of the sheep the first stages in domestication had certainly taken place by about 8920 B.C. at Zawi Chemi Shanidar in Iraq, and at Dobrudja in Rumania. But such dates and places are not necessarily definitive. The sheep may originally have been tamed by hunters in cooler regions such as the Kara Kum, east of the Caspian. Long before spinning and weaving were invented, it was possible to make wool into a warm natural felt which was invaluable in a cold climate. A single sheep, however, can eat a hundred-weight of greenstuff in a week, and the earliest herdsmen must have been kept constantly on the move in search of new grazing lands. Many would tend to move west, towards the lush pastures of Iraq, and it may have been from such nomads that the peoples of newly settled villages like Zawi Chemi Shanidar learned the techniques of sheep herding.

The pig arrived third in the barnyard, although the occasion seems to have been delayed until about 7000 B.C. One of the reasons may have been that the pig, unlike ruminant animals—goats, sheep, cattle, reindeer, and camel, for example—cannot digest straw, grass, leaves, or twigs. Pig-rearing, in fact, could not be embarked on until humans were prepared to invest some of their own food—nuts, acorns, meat scraps, cooked grain—in the enterprise.

The last major food animal to be domesticated was the cow, not apparently in the heartland areas of the neolithic, but (depending on the outcome of a current archeological controversy) either in Catal Hüyük in Turkey or Nea Nicomedia in Macedonia, at some time between 6100 B.C. and 5800 B.C. The task may have been postponed because of the difficulties involved. Though the ancestral type of the domesticated breed died out in the seventeenth century, attempts have been made in Munich and Berlin in the last twenty years to re-create it. If the fiery and agile modern version is anything to go by, neolithic man must have had his hands full with the original. . . .

Since their hunting days, humans had known that, in addition to meat, the goat provided glossy waterproof hair, and a skin which made a first-class water container; that the sheep supplied wool, and substantial quantities of fat, useful not only for cooking but as an ingredient in medicinal salves and as tallow for rushlights and lamps; that the pig's bristles were as valuable as its lard and its skin; that the cow's hide was tough and strong, and its dung an excellent fuel for the fire. But, almost certainly, it was only after the first of these animals had been domesticated that humans learned about milk and the numerous ways in which it could be used and preserved. This new foodstuff, which was to become of the greatest importance to later generations, was one of the two unlooked-for benefits of domestication. The second was that the goat, sheep and ox could be pressed into service as agricultural laborers, made to sow seeds, pull the plough, and thresh the ripe grain. The barnyard animal became, in effect, humankind's first power tool.

11. It can most reasonably be inferred from the passage that the earliest sheepherders were nomadic primarily because they:

- A. were hunters who had to travel in search of rich pastureland.
- B. were able to transport their goods in goatskin bundles.
- C. had to constantly seek new grazing land for their sheep.
- D. had to travel to mild climates more suitable for their sheep.

12. In the author's view, would the assertion that the goat was the first animal domesticated be an expression of fact or opinion, and why?

- F. Opinion, because the assertion is supported only by a balance of probabilities.
- G. Opinion, because the passage argues that the goat has been unfairly maligned.
- H. Fact, because historical research has already linked specific dates to the sheep's first domestication.
- J. Fact, because it is well known that the goat is easier to domesticate than the sheep.

13. All of the following are ruminant animals EXCEPT:
- A. sheep.
 - B. cows.
 - C. reindeer.
 - D. pigs.
14. According to the passage, the most likely reason Iraq was one of the earliest locations of sheep domestication was because:
- F. that is where the nomadic shepherders first settled.
 - G. the lush pastures there served the sheep flocks well.
 - H. the spinning and weaving of wool was invented there.
 - J. the first permanently settled villages were found there.
15. The dates referred to in the passage indicate that the domestication of the different animals mentioned occurred:
- A. over a short period of time after the domestication of wild grains.
 - B. about one or two hundred years apart from each other.
 - C. in rapid succession after the first domestication of animals.
 - D. over a long period of time, one or two thousand years apart from each other.
16. As stated in the passage, the fat of sheep has been used by humans as:
- I. a grease for cooking.
 - II. an ingredient in medicinal salves.
 - III. a fuel for the fire.
 - IV. tallow for lamps.
- F. I and II only
 - G. I, II, and IV only
 - H. II, III, and IV only
 - J. I, II, III, and IV
17. In terms of the passage as a whole, the most significant and unique aspect of the first domestication of the pig was that:
- A. humans looked upon pig-rearing as an enterprise.
 - B. humans fed foods they themselves ate to a domestic animal.
 - C. the eating of pork products was forbidden by many cultures.
 - D. the wild pig was a dangerous and intelligent animal.
18. Which of the following is the "archeological controversy" referred to in line 58?
- F. Whether the cow was first domesticated in Catal Hüyük or Nea Nicomedia
 - G. Whether the cow was the last major food animal to be domesticated
 - H. Whether the cow was first domesticated in the neolithic heartland or in Turkey or Macedonia
 - J. Whether Turkey and Macedonia are part of the neolithic heartland
19. According to the passage, which of the following was most likely the initial reason for domesticating goats and sheep?
- A. To use the animals for agricultural labor
 - B. To protect the humans' food supply of wild grain
 - C. To supply humans with a regular source of meat
 - D. To make clothing from the skins and wool of the animals
20. According to the sixth paragraph (lines 55-67), the problems involved with domesticating the cow most likely stemmed from the:
- F. fact that the ancestral breed died out in the seventeenth century.
 - G. fact that the domestic cow did not originate in the neolithic heartland.
 - H. challenge of physically controlling this temperamental animal.
 - J. need for cows to perform agricultural labor.

Passage III

HUMANITIES: This passage is adapted from Agnes de Mille's *Dance to the Piper* (©1980 by Agnes de Mille).

Anna Pavlova! My life stops as I write that name. Across the daily preoccupation of lessons, lunch boxes, tooth brushings and quarrelings with Margaret flashed this bright, unworldly experience and burned in a single afternoon a path over which I could never retrace my steps. I had witnessed the power of beauty, and in some chamber of my heart I lost forever my irresponsibility. I was as clearly marked as though she had looked me in the face and called my name. For generations my father's family had loved and served the theater. All my life I had seen actors and actresses and had heard theater jargon at the dinner table and business talk of box-office grosses. . . . But nothing in this world or my uncle's prepared me for theater as I saw it that Saturday afternoon.

Since that day I have gained some knowledge in my trade and I recognize that her technique was limited; that her arabesques were not as pure or classically correct as Markova's, that her jumps and batterie were paltry, her turns not to be compared in strength and number with the strenuous durability of Baronova or Toumanova. I know that her scenery was designed by second-rate artists, her music was on a level with restaurant orchestrations, her company definitely inferior to all the standards we insist on today, and her choreography mostly hack. And yet I say that she was in her person the quintessence of theatrical excitement.

As her little bird body revealed itself on the scene, either immobile in trembling mystery or tense in the incredible arc which was her lift, her instep stretched ahead in an arch never before seen, the tiny bones of her hands in ceaseless vibration, her face radiant, diamonds glittering under her dark hair, her little waist encased in silk, the great tutu balancing, quickening and flashing over her beating, flashing, quivering legs, every man and woman sat forward, every pulse quickened. She never appeared to rest static, some part of her trembled, vibrated, beat like a heart. Before our dazzled eyes, she flashed with the sudden sweetness of a hummingbird in action too quick for understanding by our gross utilitarian standards, in action sensed rather than seen. The movie cameras of her day could not record her allegro. Her feet and hands photographed as a blur.

Bright little bird bones, delicate bird sinews! She was all fire and steel wire. There was not an ounce of spare flesh on her skeleton, and the life force used and used her body until she died of the fever of moving, gasping for breath, much too young.

She was small, about five feet. She wore a size one and a half slipper, but her feet and hands were large in proportion to her height. Her hand could cover her whole face. Her trunk was small and stripped of all anatomy but the ciphers of adolescence, her arms and legs relatively long, the neck extraordinarily long and

mobile. All her gestures were liquid and possessed of an inner rhythm that flowed to inevitable completion with the finality of architecture or music. Her arms seemed to lift not from the elbow or the arm socket, but from the base of the spine. Her legs seemed to function from the waist. When she bent her head her whole spine moved and the motion was completed the length of the arm through the elongation of her slender hand and the quivering reaching fingers. I believe there has never been a foot like hers, slender, delicate and of such an astonishing aggressiveness when arched as to suggest the ultimate in human vitality. Without in any way being sensual, being, in fact, almost sexless, she suggested all exhilaration, gaiety and delight. She jumped, and we broke bonds with reality. We flew. We hung over the earth, spread in the air as we do in dreams, our hands turning in the air as in water—the strong forthright taut plunging leg balanced on the poised arc of the foot, the other leg stretched to the horizon like the wing of a bird. We lay balancing, quivering, turning, and all things were possible, even to us, the ordinary people.

I have seen two dancers as great or greater since, Alicia Markova and Margot Fonteyn, and many other women who have kicked higher, balanced longer or turned faster. These are poor substitutes for passion. In spite of her flimsy dances, the bald and blatant virtuosity, there was an intoxicated rapture, a focus of energy, Dionysian in its physical intensity, that I have never seen equaled by a performer in any theater of the world.

21. Which of the following descriptions most accurately and completely represents this passage?
 - A. An evocative and impassioned reminiscence of the dancer Anna Pavlova
 - B. The writer's discussion of her own dance career in relation to that of Anna Pavlova
 - C. A careful and impartial critique of the dance performances of Anna Pavlova
 - D. A comprehensive biographical sketch of Anna Pavlova's life
22. All of the following were clearly identified in the passage as dancers EXCEPT:
 - F. Toumanova.
 - G. Fonteyn.
 - H. Dionysia.
 - J. Baronova.

23. Which of the following quotations best expresses the main point of the passage?
- A. "In some chamber of my heart I lost forever my irresponsibility."
 - B. "All my life I had seen actors and actresses and had heard theater jargon at the dinner table."
 - C. "She was in her person the quintessence of theatrical excitement."
 - D. "Since that day I have gained some knowledge in my trade."
24. As it is used in line 37, the word *static* most nearly means:
- F. pulsating.
 - G. nervous.
 - H. imbalanced.
 - J. motionless.
25. It can be inferred from the passage that the writer most highly valued which of the following in a dancer?
- A. Passion and intensity
 - B. Unique physical appearance
 - C. Devotion to the theater
 - D. Classical training in dance
26. It can be most reasonably concluded from the writer's reference to "restaurant orchestrations" (line 24) that:
- F. the music typically played by restaurant orchestras was inferior.
 - G. the orchestra that played for Pavlova was made up mainly of players drawn from restaurant orchestras.
 - H. restaurant orchestras usually consisted of expert players.
 - J. Pavlova performed mostly in supper clubs and dinner theaters.
27. According to the passage, in which order did the following events occur in the writer's life?
- I. Becoming acquainted with the theater life
 - II. Seeing Markova and Fonteyn dance
 - III. Seeing Pavlova dance for the first time
- A. I, II, III
 - B. I, III, II
 - C. II, III, I
 - D. III, I, II
28. Which of the following best describes the writer's immediate reaction to seeing Pavlova for the first time?
- F. Envy of the perfect form of Pavlova's technique
 - G. Serious interest in comparing Pavlova to other dancers
 - H. Excitement at the intense beauty of Pavlova's performance
 - J. Awe and amazement at Pavlova's strength and stamina
29. All of the following aspects of Pavlova's performance are criticized in the passage EXCEPT her:
- A. choreography.
 - B. costumes.
 - C. sets and scenery.
 - D. technique.
30. The writer describes Pavlova's feet as which of the following?
- I. Slender and delicate
 - II. Suggesting vitality and aggressiveness
 - III. Large in proportion to her height
- F. I only
 - G. III only
 - H. I and II only
 - J. I, II, and III

Passage IV

NATURAL SCIENCE: This passage is adapted from an essay by David Quammen that appeared in *The Flight of the Iguana: A Sidelong View of Science and Nature* (©1988 by David Quammen).

It is one of the central axioms of ecology, held by most (though not all) scientists in the field, that the stability of an ecosystem is directly related to its complexity. The greater the number of species coexisting in one community, and the greater the number of relationships linking different species, so much greater will be the natural resistance to change, perturbation, catastrophe. From diversity comes strength; from variety, steadiness. So goes the axiom, anyway.

... But in the case of tropical coral reefs—generally judged the Earth's most complex *marine* ecosystems...—there is lately some reason for skepticism. The complexity-stability equation seems in doubt. The reason is a plague of starfish.

Acanthaster planci is the species in question, an imposing echinoderm that grows up to two feet in diameter, with as many as twenty-one arms. It moves slowly across the sea bottom in shallow Pacific waters. It feeds on live coral. Its common name is the crown-of-thorns starfish, reflecting the tangle of long sharp protective spines that protrude from the dorsal surface. Each spine is tipped with a toxic mucus. Any such two-foot-wide bush of poisonous spines is, understandably, threatened by few natural enemies.

In droves, in swarms, in startling multitudes, crown-of-thorns starfish are gobbling up the Great Barrier Reef. "There is a possibility," one scientist has written, "that we are witnessing the initial phases of extinction of madreporarian (reef-building) corals in the Pacific."

The sudden abundance of *A. planci* has raised a few interesting questions. First and most controversially: Is this plague of starfish really so bad—and so unprecedented—as it seems? Expert opinion has been divided. Second: Has the plague somehow been caused by human actions? ... Studies to prove or disprove the human role are continuing at present. Almost everyone agrees that if the starfish plague has been caused by humankind, then firm measures should be taken to control it.

The third question is rather more tricky. ... If the plague has *not* been caused by humankind—if it has been an epochal but naturally triggered catastrophe, like a lightning-caused forest fire howling across Yellowstone Park—in *that* case, should anything be done to rescue those glorious coral reefs? Or should we let unsentimental nature, and the crown-of-thorns starfish, have their way?

A. planci has been infamous to marine biologists and scuba divers for more than twenty years, since the first public reports of its population explosion at a

place called Green Island. ... Green Island marks the approximate center of that long chain of individual reefs known collectively as the Great Barrier Reef, the largest formation of coral ever seen on Earth. ...

Coral, like starfish, are animals. They belong to the same phylum as sea anemones, and in adulthood make their living as sessile creatures, attached permanently to a hard substrate, gathered together in large colonies of a particular species, each individual waving its tiny tentacles to capture planktonic food from the seawater. Their more famous attribute is that they secrete stony skeletons of calcium carbonate (lime) to support themselves. These lime skeletons—both those that still contain living coral and those that stand derelict—form the main structural matrix of coral reefs. Every reef is built up over a vast mass of compacted limestone, but it is a thin layer of live coral that keeps the reef forever renewed as a living ecosystem. That layer is precisely what *A. planci* at Green Island had begun to devour.

From 1962 to 1964 the starfish population grew inexorably. The various species of stony corals were killed off in large swaths and patches. The starfish legions accomplished this at a slow but implacable pace, each single starfish destroying roughly a square yard of coral per month. ... Behind the advancing front of starfish, which shuffled along like a herd of headless porcupines, were left dead coral skeletons that remained ghostly white for a few days or weeks, then gradually took on a gangrenous film of algae. The algae penetrated the lime skeletons and eventually weakened them, until normal wave action reduced the whole edifice to rubble. By that time, of course, the reef-dwelling fishes and other members of the coral community, dependent upon that edifice of lime for their cover, were long gone. *A. planci*, grown abundant beyond proportion, was literally collapsing its own ecosystem.

31. According to the passage, most scientists feel that if humans caused the starfish plague, then humans should:

- A. suffer the consequences.
- B. learn from their mistakes.
- C. let nature be unsentimental.
- D. remedy the problem.

32. The passage likens a forest fire caused by lightning to:

- F. a starfish plague caused by humans.
- G. a starfish plague that occurred naturally.
- H. the destruction of Yellowstone Park.
- J. live coral capturing planktonic food.

33. The irony suggested in the passage concerning the biological success of the crown-of-thorns starfish is that it is now:
- A. destroying an enormous amount of coral.
 - B. destroying its own life-support system.
 - C. ruining the skindiving in the Pacific.
 - D. ceasing to biologically diversify.
34. The passage states that coral is classified as:
- F. an animal.
 - G. a plant.
 - H. a substrate.
 - J. an echinoderm.
35. According to the passage, most scientists believe that complex, diverse ecosystems usually:
- A. have a better chance for long-term survival.
 - B. are more vulnerable to being thrown off-balance.
 - C. are younger than simple ecosystems.
 - D. lack intensely competitive species.
36. An argument presented in the passage for NOT controlling the starfish population explosion is that:
- F. the plague of *A. planci* may be the fault of human beings.
 - G. the destruction of the Great Barrier Reef will have positive effects.
 - H. human beings should try to maintain large marine ecosystems.
 - J. human beings should not interfere with nature's complex processes.
37. As it is used in line 66, the word *derelict* most nearly means:
- A. void of living organisms.
 - B. without a sense of duty.
 - C. without a real purpose.
 - D. isolated from the main structure.
38. According to the passage, live coral is necessary to preserve the reef as an ecosystem because it:
- F. belongs to the same phylum as sea anemones.
 - G. gathers together in large colonies and captures planktonic food.
 - H. forms large colonies that provide food for reef-dwelling fish.
 - J. secretes supportive lime skeletons that renew the matrix of reefs.
39. According to the passage, which of the following factors finally reduced the weakened lime skeletons in the Great Barrier Reef to rubble?
- A. The crown-of-thorns starfish
 - B. Reef-dwelling fishes
 - C. Water movement
 - D. Algae film
40. Which of the following correctly orders the stages in the destruction of a coral reef?
- I. Algae film penetrates the lime.
 - II. Dead coral skeletons turn white.
 - III. The skeletons reduce to rubble.
 - IV. Reef-dwelling fishes abandon the area.
- F. I, II, III, IV
 - G. I, II, IV, III
 - H. II, I, IV, III
 - J. II, IV, III, I

END OF TEST 3

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

DO NOT RETURN TO A PREVIOUS TEST.

SCIENCE REASONING TEST

35 Minutes—40 Questions

DIRECTIONS: There are seven passages in this test. Each passage is followed by several questions. After reading a passage, choose the best answer to each question and fill in the corresponding oval on your answer document. You may refer to the passages as often as necessary.

You are NOT permitted to use a calculator on this test.

Passage I

When two male crickets encounter each other there is either a fight or a threat display. The dominant cricket usually forces the submissive cricket to retreat to a place of safety. A biologist conducted two experiments to investigate the formation of *dominance hierarchies* (rankings) in male crickets. In the experiments described below, five adult male crickets were placed together in a container, and their interactions were observed and recorded.

Experiment 2

The crickets were placed back into the container in the same sequence as in Experiment 1. The results of all aggressive encounters between pairs of crickets were recorded. A cricket was declared a "winner" if it forced the other cricket, the "loser," to retreat. Table 2 shows the results of the interactions between the crickets.

Table 2

Winning cricket	Number of times cricket lost				
	A	B	C	D	E
A	—	5	0	20	25
B	15	—	15	20	20
C	25	25	—	10	35
D	5	0	0	—	15
E	0	0	0	10	—

Table 3 summarizes the results of all the encounters for each cricket.

Table 3

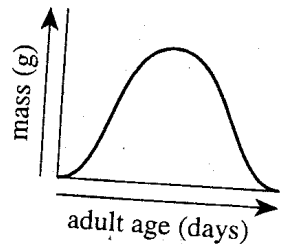
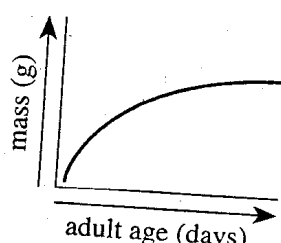
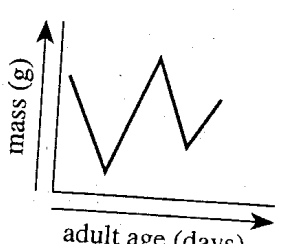
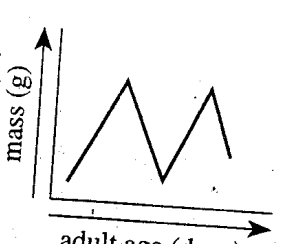
Cricket	Total number of:		
	wins	losses	encounters
A	50	45	95
B	70	30	100
C	95	15	110
D	20	60	80
E	10	95	105

Experiment 1

To determine what factors might affect the formation of dominance hierarchies, the biologist recorded the sequence in which the crickets were placed in the container, their masses, their adult ages, and the number of chirps each cricket made during the experiment. In addition, the crickets were ranked according to their aggressiveness toward each other, from most aggressive (1) to least aggressive (5). The results are shown in Table 1.

Table 1

Cricket	Sequence	Mass (grams)	Adult age (days)	Number of chirps	Aggressiveness (rank)
A	1st	0.1	2	25	3
B	2nd	0.4	1	30	2
C	3rd	0.2	4	35	1
D	4th	0.5	3	20	4
E	5th	0.3	5	0	5

1. Which of the following generalizations about the relationship between cricket mass and position in the dominance hierarchy is consistent with the experimental results?
- The smallest cricket will be the most dominant.
 - The largest cricket will be the most submissive.
 - The largest cricket will be the most dominant.
 - Weight has no effect on position in the hierarchy.
2. It was suggested that the more dominant a male cricket was, the more resources, such as food, that cricket would acquire. Accordingly, one would predict, based on win-loss records, that the cricket which would obtain the LEAST amount of food would be:
- Cricket A.
 - Cricket C.
 - Cricket D.
 - Cricket E.
3. A sixth cricket, whose mass was 0.2 grams and whose adult age was 3 days was added to the experimental container. It was observed that the cricket chirped a total of 23 times during the experiment. Based on the results of Experiment 1, what would be the rank of the sixth cricket in terms of its aggressiveness?
- 2
 - 3
 - 4
 - 5
4. According to the results of Experiments 1 and 2, which of the following factors is(are) related to the number of chirps a cricket will make?
- Mass
 - Age
 - Aggressiveness
- I only
 - III only
 - I and II only
 - I and III only
5. One can conclude from the results of Experiment 2 that Cricket D and Cricket E had a total of how many encounters with each other?
- 10
 - 15
 - 20
 - 25
6. Which of the following graphs best represents the relationship between cricket mass and age as an adult?
- F.
- 
- G.
- 
- H.
- 
- J.
- 

Passage II

Global warming can occur when solar radiation passes through the atmosphere, heats Earth's surface, and is then re-emitted as longer wavelength infrared radiation. This infrared radiation can be trapped by "greenhouse gases" in the atmosphere as it leaves Earth's surface, gradually increasing global temperatures. Greenhouse gases include water vapor, carbon dioxide (CO₂), methane, ozone, and chlorofluorocarbons.

Two scientists present their views on global warming.

Scientist 1

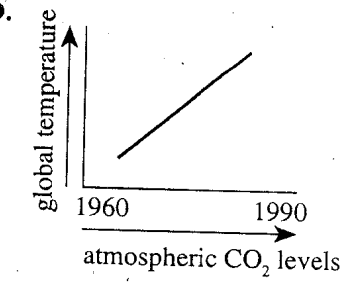
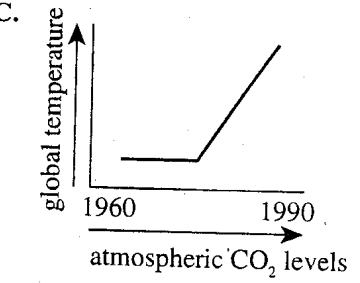
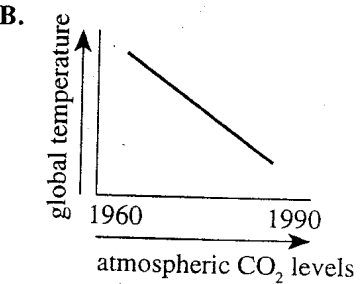
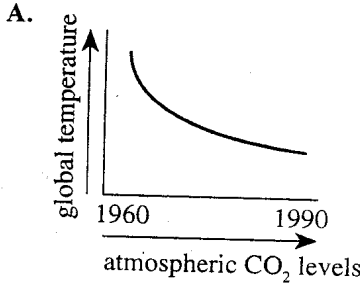
In the past 30 years, increased use of coal and oil and the burning of forests have steadily increased atmospheric CO₂. Burning coal and oil releases 5.6 billion tons of CO₂ per year; burning forests adds another 0.4 to 2.5 billion tons and also decreases the amount of plants that would remove CO₂ from the air during photosynthesis. This increase of CO₂ in the air has increased global average temperatures and caused Arctic ice to melt. One result of continued global warming will be rising sea levels due to large-scale melting of ice in polar areas.

Scientist 2

CO₂ has not caused global warming. In the past 30 years, human activity has released not only CO₂, but large amounts of dust and aerosol particles into the atmosphere. These particles reflect large amounts of incoming solar radiation back into space before it reaches Earth's surface. This actually cools Earth slightly. From 1938 to 1976, studies showed that global average temperatures were falling while CO₂ levels were increasing.

Ice ages occurred every 2,500 years and were separated by warmer periods. Even though CO₂ levels were higher during the warmer periods than during the ice ages, the ice ages occurred regardless of the CO₂ levels in the atmosphere. During the most recent warmer period 1,000 years ago, the global average temperature was 1° to 3.5° C above that of today but the CO₂ levels were much lower than today's levels. Changes in the CO₂ levels in the past were a result of the temperature changes, not a cause of them.

7. According to Scientist 1, which of the following figures best represents the global temperature and CO₂ relationship over the past 30 years?



8. Which of the following would Scientist 1 suggest to solve the problem of global warming?
- F. Reforest large areas of the planet
 - G. Increase the use of coal and oil
 - H. Reduce the number of nuclear power plants
 - J. Increase the production of chlorofluorocarbons

9. Volcanic eruptions can release large amounts of ash into the atmosphere and often cause slight global cooling. This information would best support the view of:
- A. Scientist 1, because increased CO_2 in the atmosphere causes global cooling.
 - B. Scientist 1, because increased CO_2 in the atmosphere causes global warming.
 - C. Scientist 2, because increased dust in the atmosphere causes global cooling.
 - D. Scientist 2, because increased dust in the atmosphere causes global warming.
10. According to the hypothesis of Scientist 1, which of the following results is expected if burning of forests increases?
- F. CO_2 levels in the atmosphere would decrease rapidly.
 - G. There would be a decrease in global average temperature.
 - H. The ocean levels would gradually rise.
 - J. Particle levels in the atmosphere would decrease.
11. The views of both scientists are similar because they imply that:
- A. changes in atmospheric temperature occur almost instantly after CO_2 is added.
 - B. the atmosphere's composition and average temperature remain constant over thousands of years.
 - C. the destruction of more forests could bring on an ice age sooner than expected.
 - D. there is a correlation between atmospheric temperature and materials added to the atmosphere by human activity.
12. Scientist 2 states that higher levels of:
- F. CO_2 raise global temperatures.
 - G. CO_2 do not raise global temperatures.
 - H. dust and aerosol particles raise global temperatures.
 - J. ozone do not raise global temperatures.
13. The hypothesis of Scientist 2 could best be tested by:
- A. analyzing data on wind speeds in the atmosphere for the past 50 years.
 - B. taking air temperature over the poles during the next 5 years.
 - C. measuring high-altitude wind speeds in polar areas.
 - D. measuring air temperature changes in greenhouses with different levels of airborne particles.

Passage III

Gas effusion occurs when a gas moves through a tiny opening into a region of lower pressure. The rate in milliliters per second (mL/sec) is measured using the apparatus in Figure 1.

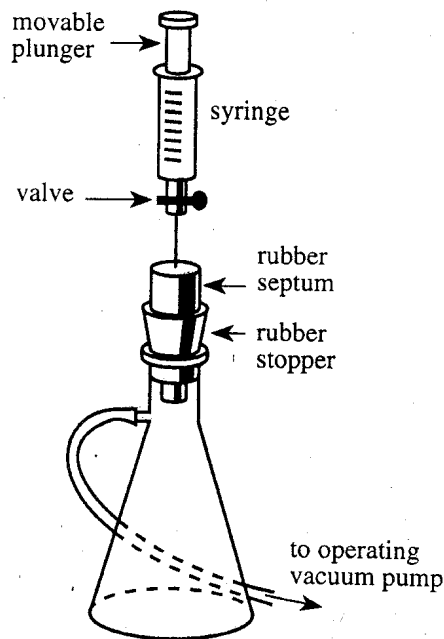


Figure 1

Table 1

Gas	Effusion time (sec)	Molecular mass (a.m.u.)*	Density (g/L)	Average molecular speed (m/sec)
H ₂	5.1	2.0	0.08	1,928
He	7.2	4.0	0.16	1,363
NH ₃	14.2	17.0	0.69	661
Air	18.2	29.0	1.19	506
O ₂	19.2	32.0	1.32	482
CO ₂	22.5	44.0	1.80	411
SO ₂	27.4	64.1	2.62	341

*1.0 a.m.u. = 1.66×10^{-27} kg (kilograms)

Figure 2 shows a graph of the effusion rate versus the molecular mass of the gases.

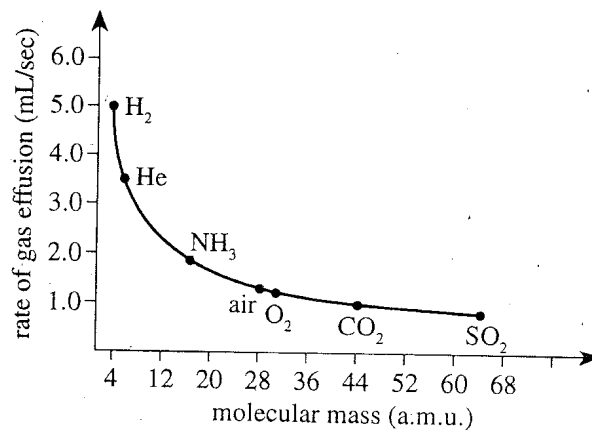


Figure 2

A 25 mL gas sample in a syringe freely escapes into a flask. The molecular mass, in a.m.u. (atomic mass units), for seven gases and the time required to completely effuse are recorded in Table 1. The densities (mass/volume) in grams per liter (g/L) and *average molecular speed* (average distance traveled by a molecule per second) in meters per second (m/sec) are also given.

14. Given the information in Table 1, the difference in effusion times is *greatest* between which of the following pairs of gases?

F. H₂ and He
 G. NH₃ and CO₂
 H. NH₃ and SO₂
 J. Air and O₂



15. What is the relationship between molecular mass and effusion time?
- A. Molecular mass decreases as effusion time increases.
 - B. Molecular mass increases as effusion time increases.
 - C. Molecular mass stays the same as effusion time increases.
 - D. Molecular mass stays the same as effusion time decreases.
16. Toy balloons are made out of a rubber material which normally has many microscopic pores. Five identical balloons are filled to a volume of 1.0 L, with CO_2 , air, NH_3 , He, and H_2 , respectively. According to the data in Table 1, if these are kept under the same conditions, which balloon will deflate first?
- F. H_2
 - G. He
 - H. NH_3
 - J. Air
17. A 25 mL sample of CO_2 gas is allowed to effuse in an apparatus like that in Figure 1. Given the information in Table 1, what percentage of the original number of CO_2 gas molecules will remain in the syringe after 11.25 sec of effusion?
- A. 12.25%
 - B. 25%
 - C. 50%
 - D. 75%
18. According to Table 1, if a syringe of unknown volume is filled with H_2 gas, and takes 15.3 sec for this H_2 gas to effuse, which of the following measurements would the unknown volume of the syringe be closest to?
- F. 6.25 mL
 - G. 12.5 mL
 - H. 50.0 mL
 - J. 75.0 mL

Passage IV

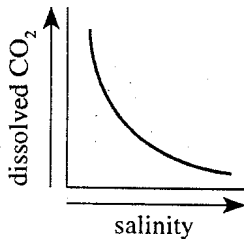
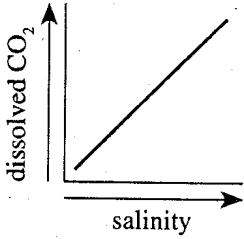
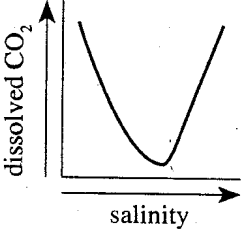
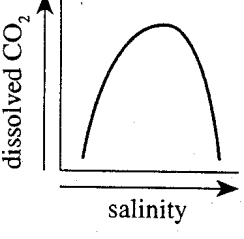
A study was conducted to determine whether two test kits (Kit A and Kit B) provided reliable data on the content of water samples of varying *salinity* (salt concentration). The concentrations of several compounds commonly tested for in water samples were measured. The results are presented in the table below. The results for the test kits were compared to estimates obtained using Standard Methods, which provide extremely accurate estimates.

Concentration (mg/L) of:	Salinity of sample				
	0 ppt*	5 ppt	10 ppt	20 ppt	30 ppt
Dissolved O ₂					
Standard	7.3	7.0	6.8	6.6	6.1
Kit A	7.3	6.9	6.6	6.4	5.9
Kit B	7.4	7.1	6.9	6.7	6.2
Dissolved CO ₂					
Standard	38.7	23.6	10.3	4.9	43.7
Kit A	41.4	30.3	9.3	7.9	41.8
Kit B	36.0	22.1	10.5	11.0	44.0
Chloride					
Standard	26.9	1,350	4,568	9,138	17,924
Kit A	28.9	1,500	5,230	9,870	18,220
Kit B	46.4	2,130	6,832	13,500	27,750
CaCO ₃					
Standard	70.1	72.3	75.9	79.0	123.4
Kit A	68.0	70.0	73.3	77.8	124.2
Kit B	75.1	76.5	79.8	83.2	127.4
NH ₃					
Standard	0.32	0.41	0.55	0.65	0.60
Kit A	0.58	0.71	0.79	0.88	0.84
Kit B	0.29	0.39	0.53	0.68	0.64

* ppt = parts per thousand

Note: Each estimate is the average of 5 measurements.

Table adapted from Claude E. Boyd and Harry V. Daniels, "Evaluation of Hach Fish Farmer's Water Quality Test Kits for Saline Water." ©1988 by the World Aquaculture Society.

19. From the results, one would conclude that at a salinity of 0 ppt, Kit A is most accurate in measuring the concentration of which of the following compounds, relative to the Standard Method?
- Dissolved O_2
 - Dissolved CO_2
 - Chloride
 - $CaCO_3$
20. The data from which of the measurement procedures supports the conclusion that the concentration of chloride increases as the level of salinity increases?
- Standard Method only
 - Kit A only
 - Standard Method and Kit A only
 - Standard Method, Kit A, and Kit B
21. Is the conclusion that Kit A is more accurate than Kit B for estimation of NH_3 concentration supported by the results in the table?
- Yes, because the estimates using Kit A are consistently lower than the estimates using Kit B.
 - Yes, because the estimates using Kit A are more similar to the estimates using the Standard Method than the estimates using Kit B.
 - No, because the estimates using Kit B are consistently higher than the estimates using Kit A.
 - No, because the estimates using Kit B are more similar to the estimates using the Standard Method than the estimates using Kit A.
22. Which of the following graphs best represents the relationship between the level of salinity and the concentration of dissolved CO_2 , as estimated by Kit A?
- F.
- 
- G.
- 
- H.
- 
- J.
- 
23. A water sample of unknown salinity was tested using Kit A. The concentrations, in mg/L, of selected compounds in this sample were: dissolved $O_2 = 7.1$, $CaCO_3 = 69.0$, and $NH_3 = 0.65$. According to the data in the table, one would predict that the most likely level of salinity was:
- 0 ppt.
 - 3 ppt.
 - 15 ppt.
 - 25 ppt.

Passage V

Moving water can shape sediment into *bedforms*, such as *ripples* (narrow ridges separated by wider troughs). A geologist performed 3 studies to examine bedforms.

Study 1

Samples were taken from 4 layers of an exposed section of sandstone and the average sample grain diameter, in millimeters (mm), was measured. Any bedforms preserved in the sections were recorded. The results are shown in Table 1.

Table 1

Section	Average grain diameter (mm)	Bedform
4 (top)	0.2	small ripples
3	0.5	large ripples
2	0.5	flat bed
1 (bottom)	10	none

Study 2

A sediment sample was taken at 5 sites across the stream shown in Figure 1. The average grain diameter of each sample was determined and any bedforms at the site were noted. The water depth and velocity, in centimeters per second (cm/sec), were measured at Sites 3, 4, and 5. The results are shown in Table 2.

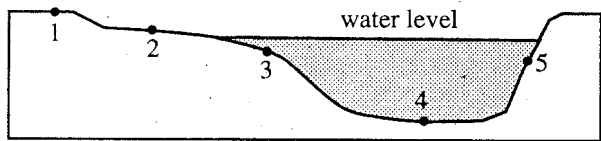


Figure 1

Table 2

Site	Water depth (cm)	Water velocity (cm/sec)	Average grain diameter (mm)	Bedform
1	—	—	0.05	none
2	—	—	0.5	small ripples
3	20	5	0.5	large ripples
4	100	15	10	large ripples
5	10	10	0.05	none

Study 3

A stream channel was built that contained 20 cm deep flowing water. Three experiments were done, each using a flat layer of a different sediment grain diameter. The bedforms created by flowing water of various velocities were recorded for each experiment. The results are shown in Table 3. “Flat bed” means sediment movement but no ripples. “NM” means no sediment movement.

Table 3

Grain diameter (mm)	Bedforms at a water velocity (cm/sec) of:			
	10	40	80	120
0.05 (silt)	small ripples	small ripples	flat bed	sediment completely removed
0.5 (sand)	NM	small ripples	large ripples	flat bed
2.0 (gravel)	NM	flat bed	large ripples	large ripples

24. If *gravel* is any sediment with a grain diameter of 2 mm or greater, what site in Study 2 had gravel sediment?

- F. Site 1
- G. Site 2
- H. Site 3
- J. Site 4

25. According to Study 3, sand would most likely be completely washed away by water moving at a velocity of:

- A. between 10 and 40 cm/sec.
- B. between 40 and 80 cm/sec.
- C. between 80 and 120 cm/sec.
- D. over 120 cm/sec.



26. Which of the following statements best describes the purpose of Studies 2 and 3 ?

- F. To demonstrate conditions that could create bedforms similar to those found in the sandstone outcrop
- G. To determine the cross-sectional shape of the stream that originally deposited the sand found in the outcrop
- H. To measure the volume of water in a small river
- J. To identify the kinds of plants that lived at the time the sand was being changed to sandstone

27. Closer examination of Section 1 in Study 1 revealed that the largest grains (up to 20 mm in diameter) were at the bottom of the section. According to the studies, which of the following statements best explains why such large grains were found at the bottom of the section?

- A. The water velocity at the time the sediment was deposited was high enough to carry away all but the largest grains.
- B. The water velocity at the time the sediment was deposited was too slow to carry away even the smallest grains.
- C. The water depth at the time the sediment was deposited was too shallow for large grains to settle to the bottom.
- D. The water depth at the time the sediment was deposited was too deep for large grains to settle to the bottom.

28. According to Study 3, which of the following is the best explanation for the presence of ripples at Site 2 in Study 2 ?

- F. Site 2 sediment once had an average grain diameter of 10 mm.
- G. Water moving at a velocity of 40 cm/sec once covered Site 2.
- H. Water moving at a velocity of 120 cm/sec once covered Site 2.
- J. Water 100 cm deep once covered Site 2.

29. Another sample was taken from a sixth site in Study 2 that yielded the following data:

Site	Water depth (cm)	Water velocity (cm/sec)	Average grain diameter (mm)	Bedform
6	50	5	0.5	large ripples

According to this data and the data from Study 2, one could generalize that:

- A. large ripples only form in sediments having a diameter of 0.5 mm.
- B. large ripples only form in sediments having a diameter of 10 mm.
- C. water depth has no influence on the type of bedform produced.
- D. water depth must be less than 50 cm before bedforms will be created by moving water.

Passage VI

Light causes some metals to eject electrons. The energy of those electrons, in joules (J), depends on the light frequency and the kind of metal. Figure 1 shows how the maximum energy of ejected electrons varies with frequency, in hertz (Hz), for 5 metals. Figure 2 shows how the number of electrons emitted by a metal varies with the intensity of light at a constant frequency. Figure 3 shows the maximum energy an ejected electron can have for a given intensity of light at a constant frequency.

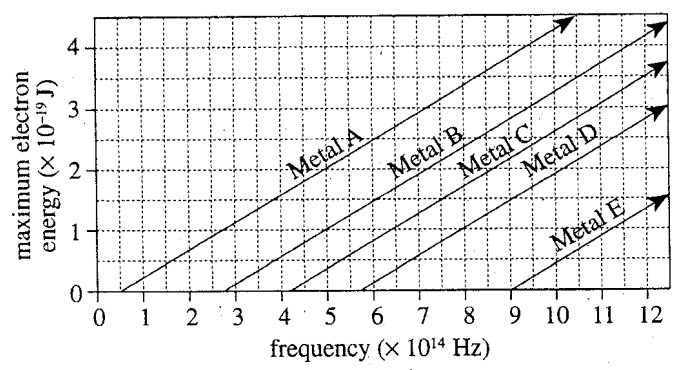


Figure 1

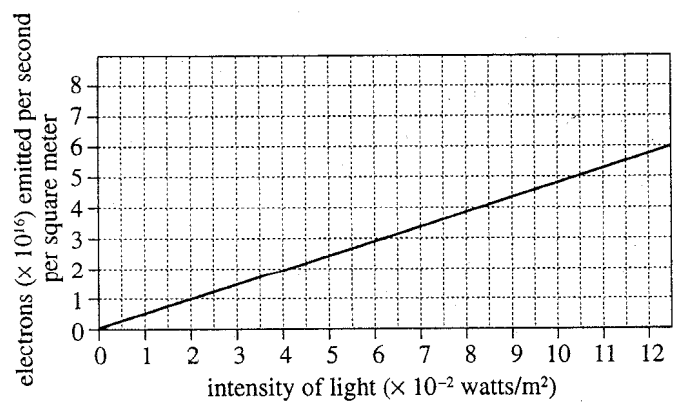


Figure 2

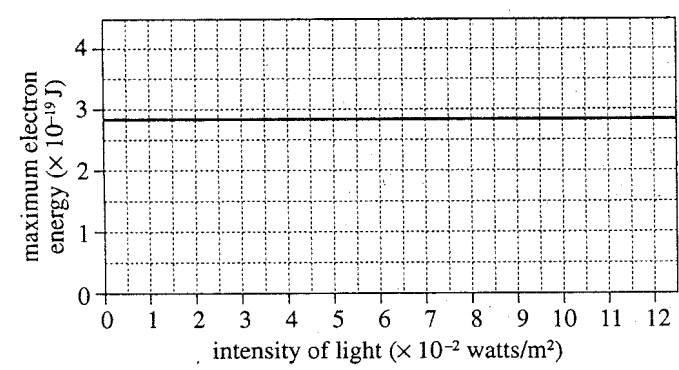


Figure 3

30. Which of the following data is represented on the x-axis of Figure 2 ?
- F. Frequency
 - G. Intensity of light
 - H. Number of electrons emitted per second
 - J. Maximum energy of emitted electrons
31. The hypothesis that turning on more light will increase the number of electrons emitted per second is supported by which figure(s)?
- (Note: Assume the additional light also has adequate energy to eject electrons from metals.)
- A. Figure 1 only
 - B. Figure 2 only
 - C. Figure 3 only
 - D. Figures 2 and 3 only
32. Is the statement "The maximum energy of electrons increases with intensity" supported by the data in Figure 3 ?
- F. Yes, because the graph line moves to the right as the value of intensity increases.
 - G. Yes, because the maximum energy of electrons increases as the value of intensity increases.
 - H. No, because the same value for maximum energy is indicated for all values of intensity.
 - J. No, because the line does not slope downward from left to right.
33. If the data in Figure 3 was produced by a light with a frequency of 6.5×10^{14} Hz, which metal was most likely used?
- A. Metal A
 - B. Metal B
 - C. Metal C
 - D. Metal E

34. According to the data, which of the following statements best compares the frequencies of light required for both Metals A and D to produce an electron with a maximum electron energy of $2 \times 10^{-19} \text{ J}$?

- F. The frequency required for Metal D is approximately one-quarter of that required for Metal A.
- G. The frequency required for Metal D is approximately half of that required for Metal A.
- H. The frequency required for Metal D is the same as that required for Metal A.
- J. The frequency required for Metal D is approximately twice that required for Metal A.

Passage VII

A student conducted experiments to determine the relationship between the *musical pitch* (frequency) of a vibrating string and its length, tension, and *mass density* (mass per unit length of string).

Experiment 1

The student cut several lengths of steel strings having different diameters. The strings were weighed and their mass densities calculated. Specified lengths of the strings were hung from the ceiling. A block was suspended from the free end of each string, making the tension in the string equal to the suspended weight in newtons (N). Plucking the string caused it to vibrate, producing sound waves. The sound waves were detected by a microphone and displayed on the screen of an oscilloscope, where the fundamental *frequency of vibration*, in cycles per second, or hertz (Hz), could be measured. The results are presented in Table 1.

Table 1

Trial	Vibrating length (m)	Mass density (g/m)	Weight suspended (N)	Frequency (Hz)
1	1.0	10	900	150
2	1.0	10	3,600	300
3	1.0	40	900	75
4	1.0	40	3,600	150
5	0.5	10	900	300
6	0.5	10	3,600	600
7	0.5	40	900	150

Experiment 2

The student repeated the procedure in Experiment 1, using strings made of nylon and gut and having different diameters. The results are presented in Table 2.

Table 2

Trial	Material	Vibrating length (m)	Mass density (g/m)	Weight suspended (N)	Frequency (Hz)
8	nylon	1.0	40.0	900	75
9	nylon	1.0	10.0	3,600	300
10	nylon	1.0	2.5	900	300
11	nylon	1.0	2.5	225	150
12	gut	1.0	10.0	900	150
13	gut	1.0	2.5	3,600	600

35. The variable that is directly manipulated by the student between Trials 10 and 11 is the:
- mass density of the string.
 - frequency of the resulting vibration.
 - material of which the string is composed.
 - tension to which the string is subjected.
36. Instead of attaching a block to the string in Trial 1, the student could have produced the same tension in the string by:
- increasing the length of the string to 3.0 m.
 - pulling on the string with a force of 900 N.
 - increasing the mass density to 20 g/m.
 - plucking the string close to the ceiling.
37. If the string used in Trial 10 were subjected to a tension of 3,600 N, the frequency of the resulting sound waves would be:
- 37 Hz.
 - 75 Hz.
 - 150 Hz.
 - 600 Hz.
38. On the basis of both experiments, which of the following statements regarding the relationship between observed frequency of vibration and the other physical variables is correct?
- Increasing tension increases observed frequency.
 - Increasing mass density increases observed frequency.
 - Increasing vibrating length increases observed frequency.
 - The observed frequency depends on the material.



39. A student is given several nylon strings of identical vibrating length but unknown mass densities. The string with the highest mass density could be found by:
- A. placing all the strings under identical tension and selecting the string with the highest frequency.
 - B. placing all the strings under identical tension and selecting the string with the lowest frequency.
 - C. tuning all the strings to the same frequency and selecting the string with the lowest tension.
 - D. selecting the string capable of sustaining the highest frequency without breaking.
40. On the Diatonic C major scale, a middle D note is taken to be 297 Hz. A guitarist wants to use a steel string of mass density 40 g/m with a tension of 3,600 N for the D string on a special type of guitar. Approximately what length of string should the guitarist use to produce a middle D note?
- F. 0.5 m
 - G. 1.5 m
 - H. 2.0 m
 - J. 3.0 m

END OF TEST 4

STOP! DO NOT TURN THE PAGE UNTIL TOLD TO DO SO.

DO NOT RETURN TO A PREVIOUS TEST.

ACT Resource Links

ACT Online Practice Tests: <https://www.crackab.com/act/>

✧ **ACT English Practice Tests:**

<https://www.crackab.com/act/english/>

✧ **ACT Math Practice Tests:**

<https://www.crackab.com/act/math/>

✧ **ACT Reading Practice Tests:**

<https://www.crackab.com/act/reading/>

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<https://www.apstudy.net>

Scoring Keys for the ACT Sample Test Booklet 9954D

Use the scoring key for each test to score your answer document for the sample test. Mark a "1" in the blank for each question you answered correctly. Add up the numbers in each subscore area and enter the total number correct for each subscore area in the blanks provided. Also enter the total number correct for each test in the blanks provided. The total number correct for each test is the sum of the number correct in each subscore area.

Test 1: English—Scoring Key

Subscore Area*			Subscore Area*			Subscore Area*		
Key	UM	RH	Key	UM	RH	Key	UM	RH
1. B			26. J			51. D		
2. G			27. A			52. F		
3. B			28. H			53. C		
4. H			29. A			54. G		
5. C			30. F			55. D		
6. J			31. C			56. J		
7. C			32. J			57. C		
8. G			33. A			58. F		
9. D			34. H			59. A		
10. F			35. B			60. F		
11. D			36. J			61. C		
12. H			37. D			62. H		
13. B			38. G			63. D		
14. F			39. D			64. H		
15. D			40. J			65. B		
16. G			41. A			66. G		
17. C			42. G			67. A		
18. J			43. C			68. H		
19. B			44. G			69. A		
20. H			45. D			70. G		
21. A			46. H			71. A		
22. F			47. D			72. G		
23. A			48. F			73. C		
24. G			49. C			74. H		
25. D			50. F			75. C		

Number Correct (Raw Score) for:

Usage/Mechanics (UM) Subscore Area	(40)
Rhetorical Skills (RH) Subscore Area	(35)
Total Number Correct for English Test (UM + RH)	(75)

* UM = Usage/Mechanics
RH = Rhetorical Skills

Test 2: Mathematics—Scoring Key

	Key	Subscore Area*		
		EA	AG	GT
1.	C	_____		
2.	F	_____		_____
3.	C	_____		
4.	K	_____		
5.	D	_____		
6.	F	_____		
7.	A	_____		_____
8.	H	_____		
9.	B	_____		
10.	F	_____		
11.	D	_____		
12.	K	_____	_____	
13.	E	_____		_____
14.	K	_____	_____	
15.	A	_____		
16.	G	_____		
17.	C	_____		
18.	H	_____		
19.	D	_____		
20.	J	_____	_____	
21.	A	_____		
22.	K	_____		_____
23.	C	_____		
24.	G	_____		
25.	E	_____		
26.	J	_____	_____	
27.	B	_____	_____	
28.	F	_____	_____	
29.	D	_____	_____	
30.	K	_____		_____

	Key	Subscore Area*		
		EA	AG	GT
31.	B		_____	
32.	J		_____	_____
33.	C			_____
34.	J		_____	
35.	C			_____
36.	F		_____	
37.	C	_____		
38.	G		_____	
39.	C		_____	
40.	J			_____
41.	E	_____		
42.	H	_____		
43.	B			_____
44.	F			_____
45.	C			_____
46.	G		_____	
47.	B		_____	
48.	K			_____
49.	B			_____
50.	H			_____
51.	D			_____
52.	K		_____	
53.	A		_____	
54.	G		_____	
55.	A			_____
56.	K	_____		
57.	D	_____		
58.	J			_____
59.	B		_____	
60.	G	_____		

Number Correct (Raw Score) for:

Pre-Alg./Elem. Alg. (EA) Subscore Area

_____ (24)

Inter. Alg./Coord. Geo. (AG) Subscore Area

_____ (18)

Plane Geo./Trig. (GT) Subscore Area

_____ (18)

Total Number Correct for Math Test (EA + AG + GT)

_____ (60)

* EA = Pre-Algebra/Elementary Algebra

AG = Intermediate Algebra/Coordinate Geometry

GT = Plane Geometry/Trigonometry

Test 3: Reading—Scoring Key

Subscore Area*			Subscore Area*			Subscore Area*		
Key	SS	AL	Key	SS	AL	Key	SS	AL
1. C			15. D			29. B		
2. F			16. G			30. J		
3. A			17. B			31. D		
4. J			18. F			32. G		
5. D			19. B			33. B		
6. H			20. H			34. F		
7. B			21. A			35. A		
8. J			22. H			36. J		
9. C			23. C			37. A		
10. G			24. J			38. J		
11. C			25. A			39. C		
12. F			26. F			40. H		
13. D			27. B					
14. G			28. H					

Number Correct (Raw Score) for:

Social Studies/Sciences (SS) Subscore Area

(20)

Arts/Literature (AL) Subscore Area

(20)

Total Number Correct for Reading Test (SS + AL)

(40)

* SS = Social Studies/Sciences

AL = Arts/Literature

Test 4: Science Reasoning—Scoring Key

Key		Key		Key	
1. D		15. B		29. C	
2. J		16. F		30. G	
3. C		17. C		31. B	
4. G		18. J		32. H	
5. D		19. A		33. A	
6. H		20. J		34. J	
7. D		21. D		35. D	
8. F		22. H		36. G	
9. C		23. B		37. D	
10. H		24. J		38. F	
11. D		25. D		39. B	
12. G		26. F		40. F	
13. D		27. A			
14. H		28. G			

Number Correct (Raw Score) for:

Total Number Correct for Science Reasoning Test

(40)